To my family, and to everyone who discussed a factor graph with me at improbable hours across improbable time zones.

Contents

Chapter 1

Introduction

1.1 The problem

Generative diffusion models synthesise data by reversing a noising process, and the object that makes the reversal possible is the *score function*, the gradient of the log-density of progressively noised data,

$$S(x,t) \;=\; \nabla_x \log p_t(x) \tag{1.1}$$

(Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021; Anderson, 1982). In applications the score is approximated by a neural network trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011); the network is a black box, and the structure of the function it has to learn is hard to inspect. A complementary research programme, developed in the statistical-physics community, studies data distributions simple enough that the score can be computed exactly, yet rich enough to reproduce phenomena observed in real systems (Biroli and Mézard, 2023; Biroli et al., 2024; Raya and Ambrogioni, 2023; Ghio et al., 2024). Exact scores make every regime and transition along the generation trajectory traceable to explicit formulas.

The models analysed in that programme treat a data point as an unstructured collection of coordinates. Much real data is not of this kind: a video, an audio signal, or a time series is a *trajectory*, a sequence of frames coupled by a dynamical rule. For such data the relevant score is the *joint* score of the whole noisy sequence, and its structure is shaped by the interaction of two times: the internal (frame) time of the sequence and the diffusion time of the noising process. This thesis studies that interaction in the simplest setting in which it is nontrivial and exactly solvable.

1.2 The model and the gap

The data model is the stationary first-order autoregressive chain, $a_{k+1} = \alpha a_k + \eta_k$ with $|\alpha| < 1$ and $\sigma_\eta^2 = 1 - \alpha^2$, whose trajectory $a = (a_0, \ldots, a_{K-1})$ constitutes one data point. Each frame is corrupted independently by the Ornstein–Uhlenbeck channel for a diffusion time t. Two classical bodies of theory meet here. The joint score of the noisy sequence is governed by the algebra of structured matrices: Toeplitz covariances, tridiagonal precisions, and their common eigenbasis. And, by an identity that holds for any data distribution, the score is an affine function of a posterior expectation, so computing it is Bayesian smoothing on a chain, the domain of belief propagation and of the Kalman filter (Kalman, 1960; Rauch et al., 1965; Pearl, 1988; Kschischang et al., 2001).

Three gaps motivate the work (the supporting literature is reviewed in Chapter 2). The exactly solvable diffusion literature does not treat data points with internal Markov structure. The classical smoothing literature fixes the observation noise and does not ask how the posterior, hence the score, deforms as the diffusion time varies. And the Gaussian message approximations of applied inference are classically justified by a central-limit argument that requires many incoming messages per node, whereas a chain provides two; we are not aware of a quantitative assessment of that approximation on chain-structured diffusion posteriors.

1.3 Research questions

RQ1. For a stationary Gaussian AR(1) chain observed through coordinatewise Ornstein–Uhlenbeck noise, what is the joint score, and how do its spatial couplings vary with the diffusion time?

RQ2. How can the same Gaussian-chain score be computed by belief propagation, and how does the resulting recursion relate to Kalman filtering and smoothing?

RQ3. For selected non-Gaussian innovation models, how accurate is a Gaussian-message approximation relative to a validated numerical belief-propagation reference, over the parameter ranges studied?

RQ4. In the Gaussian model, how rapidly does the influence of remote observations on the score decay, and what limited guidance does this provide for local approximations of the score?

These questions are deliberately confined to the models named in them. No claim is made about general Markov processes, about universal properties of learned diffusion scores, or about practical neural architectures.

1.4 Contributions

1. For the stationary Gaussian AR(1) chain under OU corruption, we derive the joint score in closed form, diagonalise it for all diffusion times in a single eigenbasis, and characterise the evolution of its coupling structure: a band-fill law $(Q_t)_{i,i+d} = (-1)^{d-1}(2t)^{d-1}(Q_0^d)_{i,i+d} + O(t^d)$ for small t, a return to isotropy at rate e^{-2t}, and exact bulk formulas for the posterior variance and the geometric decay rate $q(\alpha, t)$ of posterior correlations. The band-fill law corrects a prefactor error in an earlier working note of this project; the correction is verified numerically (Chapter 3).

2. We derive the belief-propagation computation of the same score on the chain's factor graph, show algebraically that Gaussian messages are closed under the updates,

identify the resulting recursion, update by update, with the Kalman filter and the Rauch–Tung–Striebel smoother, and verify that it reproduces the matrix score to within double-precision rounding across a parameter grid. We state the complexity honestly: the message-update count is $O(K)$ for any innovation law; the cost is $O(K)$ only because the Gaussian family is finite-dimensional (Chapter 3).

3. For Laplace-innovation chains we establish which identities remain exact, locate the precise step at which closed forms stop, and measure the accuracy of Gaussian-projected BP against a numerically validated grid implementation of the functional recursion: the median relative score error is 0.39 at $t = 0.02$ and decays below 10^{-3} for $t \gtrsim 0.9$, with the score direction substantially more accurate than its magnitude, in the tested regimes (Chapter 4).

4. In the Gaussian model we prove that the error of a radius-r local estimator of the score decays as $q(\alpha, t)^r$, and we measure that truncating the inference (windowed message passing) is more accurate than truncating the score matrix (banding) at small diffusion times (Chapter 3).

5. As a separate, self-contained result, we characterise the per-node (AMP/TAP) closure on the same model: it returns the exact score but a different posterior variance, with an existence boundary in closed form, a breakdown time $t_c(\alpha)$, and a critical coupling $\alpha_c = \sqrt{2} - 1$. All formulas are independently re-verified; the material is placed in Appendix B because it branches off the main line (statement of results in Section 5.1).

As a methodological point, no neural network is trained anywhere: every claim is either derived by explicit algebra or measured numerically against an independently computed reference, and all numbers in the thesis are regenerated by deterministic audit scripts (Appendix C). We regard this reproducibility as a strength of the method, not as a contribution in itself.

1.5 Scope and limitations

The exact results concern one-dimensional state per frame, linear AR(1) dynamics, and stationary initial conditions. The non-Gaussian results are numerical, valid within the validated resolution regime of the reference solver and the tested parameter ranges. The locality results are proved for the Gaussian bulk only. No learning experiments are performed, so any implication for trained score networks is a hypothesis suggested by a controlled model, not an established property of practical systems.

1.6 Outline

Chapter 2 defines the model, derives the score–posterior identity, reviews the relevant literature, and records how the research questions took shape, including the preliminary experiments. Chapter 3 contains the complete Gaussian analysis: the exact score, its spectral form and lifecycle, belief propagation and its identification with Kalman smoothing, and the locality laws. Chapter 4 contains the complete non-Gaussian analysis: what survives with Laplace innovations, where closed forms stop, and the measured cost of the Gaussian message family. Chapter 5 states findings, limitations, and future questions. The appendices collect Gaussian identities, the AMP/TAP analysis, and the reproducibility protocol.

Chapter 2

Model, Score Identities, and Research Context

This chapter contains everything the research chapters depend on: the data model (a stationary Gaussian, later Laplace, AR(1) chain), the corruption model (the Ornstein–Uhlenbeck channel), the identity that converts score computation into Bayesian inference, the strands of literature the thesis builds on, and the preliminary work that shaped the final research questions. Background appears only where a later definition, derivation, or experiment uses it; for the surrounding theory we cite standard references.

2.1 The data model: a stationary AR(1) chain

A sequence $(X_0, \dots, X_{K-1})$ is a *Markov chain* if each variable depends on the past only through its immediate predecessor; its joint density then factorises into local terms,

$$p(x_0, \dots, x_{K-1}) \;=\; p_0(x_0) \prod_{k=0}^{K-2} M(x_{k+1} \,|\, x_k), \tag{2.1}$$

with M the transition kernel (Robert and Casella, 2004). The factorisation (2.1) is the structural fact the whole thesis exploits: it makes the precision matrix of a Gaussian chain tridiagonal (Chapter 3) and the factor graph of the posterior a tree, on which belief propagation is exact.

The specific chain we study is the first-order autoregressive process, the canonical Gaussian Markov chain (Brockwell and Davis, 1991). For $\alpha \in (-1, 1)$ and i.i.d. innovations $\eta_k \sim \mathcal{N}(0, \sigma_\eta^2)$,

$$a_{k+1} \;=\; \alpha\, a_k + \eta_k, \qquad k = 0, 1, \ldots, K-2, \tag{2.2}$$

so $M(a'|a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$. Throughout we use the stationary normalisation $\sigma_\eta^2 = 1 - \alpha^2$ and start the chain in its stationary law, $a_0 \sim \mathcal{N}(0, 1)$; every frame then has unit marginal variance. A trajectory $a = (a_0, \ldots, a_{K-1})$ is *one* data point, the prototype of a video or a time series. The frame index k is an internal time, to be kept distinct from the diffusion time t introduced next.

Toolbox 2.1 — Stationary variance and autocovariance of AR(1)

(i) Stationary variance. If $a_k \sim \mathcal{N}(0, \sigma^2)$, then by independence of η_k, $\mathrm{Var}(a_{k+1}) = \alpha^2\sigma^2 + \sigma_\eta^2$. Imposing equality of the two variances gives $\sigma_\infty^2 = \sigma_\eta^2/(1-\alpha^2)$, which equals 1 under the stationary normalisation. The map $\sigma^2 \mapsto \alpha^2\sigma^2 + \sigma_\eta^2$ is a contraction for $|\alpha| < 1$, so any initial variance converges to σ_∞^2 geometrically, and Gaussianity is preserved by the affine recursion.
(ii) Autocovariance. Unrolling (2.2) d times, $a_{k+d} = \alpha^d a_k + \sum_{j=0}^{d-1} \alpha^j \eta_{k+d-1-j}$, and all innovation terms are independent of a_k, so in the stationary regime

$$\mathrm{Cov}(a_i, a_j) \;=\; \alpha^{|i-j|}.$$

Correlations decay geometrically with the lag, and the covariance matrix of K consecutive frames is symmetric Toeplitz: constant along diagonals, because the process is stationary. □

2.2 The corruption model: the Ornstein–Uhlenbeck channel

Diffusion models corrupt data with a fixed noising process. The standard choice (Song et al., 2021) is the Ornstein–Uhlenbeck (OU) process, Brownian motion pulled back to the origin by a linear spring, $\mathrm{d}X_t = -X_t\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t$ (Uhlenbeck and Ornstein, 1930). Its transition kernel is Gaussian and available in closed form (Gardiner, 2009): corrupting a value a for a time t produces

$$x \;=\; a\,e^{-t} \;+\; \sqrt{\Delta_t}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,1), \qquad \boxed{\Delta_t \;=\; 1 - e^{-2t}}. \tag{2.3}$$

The signal is contracted by e^{-t} while isotropic noise of variance Δ_t fills in: at $t = 0$ the data are untouched; as $t \to \infty$ the output is $\mathcal{N}(0,1)$ regardless of the input. The signal-to-noise ratio e^{-2t}/Δ_t sweeps from ∞ to 0, and it diverges as $1/2t$ for small t, a factor that reappears throughout as an error amplifier. Two consequences are used constantly. First, read as a function of the *clean* variable, the kernel is again Gaussian, $\mathcal{N}(x; ae^{-t}, \Delta_t) \propto \exp[-(e^{-2t}/2\Delta_t)(a - xe^t)^2]$: a Gaussian likelihood factor of precision e^{-2t}/Δ_t centred at the deconvolved observation $e^t x$. Second, pushing a Gaussian vector $a \sim \mathcal{N}(0, \Sigma_0)$ through (2.3) coordinatewise with independent noises gives a Gaussian output with

$$\Sigma_t \;=\; e^{-2t}\,\Sigma_0 \;+\; \Delta_t\, I, \tag{2.4}$$

since means and covariances transform linearly and the added noise is white. Equation (2.4) is the entire forward process of this thesis in one line.

2.3 Diffusion models and the score

Applying the channel (2.3) to data $a \sim p_0$ produces the noised marginals

$$p_t(x) \;=\; \int p_0(a)\,\mathcal{N}\Big(x;\, ae^{-t},\, \Delta_t I\Big)\,\mathrm{d}a, \tag{2.5}$$

which interpolate from the data distribution to $\mathcal{N}(0, I)$. By Anderson's time-reversal theorem (Anderson, 1982), the corruption is undone by a reverse-time SDE whose drift contains a single unknown object, the *score* of the noised marginals:

$$\mathrm{d}X_t \;=\; \Big[-X_t \;-\; 2\,\nabla_x \log p_t(X_t)\Big]\,\mathrm{d}t \;+\; \sqrt{2}\,\mathrm{d}\bar{W}_t, \qquad S(x,t) := \nabla_x \log p_t(x). \tag{2.6}$$

Whoever knows $S(x,t)$ for all t can sample from p_0; the same score also drives a deterministic transport, the probability-flow ODE (Song et al., 2021), which connects diffusion models to normalizing flows and flow matching (Chen et al., 2018; Lipman et al., 2023; Albergo and Vanden-Eijnden, 2023). In practice the score is approximated by a neural network trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011; Ho et al., 2020): one regresses a network $s_\theta(x,t)$ onto the kernel score $\nabla_x \log \mathcal{N}(x; ae^{-t}, \Delta_t I) = -(x - ae^{-t})/\Delta_t$ over pairs (a, x), which is equivalent to predicting the added noise. The score is also what makes the energy-based view of generative modelling (LeCun et al., 2006) tractable: for a model $p_\theta \propto e^{-E_\theta}$ the partition function obstructs likelihood, sampling, and learning alike, but $\nabla_x \log p_\theta = -\nabla_x E_\theta$ is free of it. Diffusion models work entirely with scores and thereby sidestep normalisation.

2.4 The score as a posterior expectation

One identity converts score computation into Bayesian inference, and every result of this thesis passes through it. Differentiating (2.5) under the integral sign,

$$\nabla_x \log p_t(x) = \frac{\int p_0(a)\,\nabla_x \mathcal{N}(x; ae^{-t}, \Delta_t I)\,\mathrm{d}a}{p_t(x)} = \int p(a \,|\, x)\,\frac{ae^{-t} - x}{\Delta_t}\,\mathrm{d}a,$$

where $p(a|x) = p_0(a)\mathcal{N}(x; ae^{-t}, \Delta_t I)/p_t(x)$ is the posterior of the clean data given the noisy observation. Hence

$$\boxed{S(x,t) \;=\; \frac{e^{-t}\,\mathbb{E}[\,a \,|\, x\,] - x}{\Delta_t}.} \tag{2.7}$$

We call (2.7) the *score–posterior identity*; in empirical Bayes it is known as Tweedie's formula (Robbins, 1956; Miyasawa, 1961; Efron, 2011). It holds for *any* data distribution p_0: the score of the noisy marginal and the Bayes-optimal denoiser $\mathbb{E}[a|x]$ determine each other exactly. Estimating a score is therefore solving an inference problem, and the quality of any approximate score is the quality of an approximate posterior mean, amplified by e^{-t}/Δ_t.

For sequence data $a \in \mathbb{R}^K$ noised coordinatewise, the identity holds componentwise with the *full* posterior:

$$S_k(x,t) \;=\; \frac{e^{-t}\,\mathbb{E}[\,a_k \,|\, x_0,\ldots,x_{K-1}\,] - x_k}{\Delta_t}. \tag{2.8}$$

The conditioning is on all frames: although the noise is independent across frames, the prior couples them, so the k-th score component depends on every observation. This *joint* score is a different object from the per-frame marginal score $\partial_{x_k} \log p_t(x_k)$, and only the joint score drives the reverse dynamics (2.6) of the sequence as a whole. For our stationary model the distinction is extreme: the marginal score is degenerate (Section 2.6), while the joint score carries all the structure. When the prior is a Markov chain, the posterior in (2.8) is a smoothing distribution on a chain, which is what makes exact computation possible.

2.5 Related work

Statistical physics of learning and of diffusion. The statistical-mechanics analysis of neural computation runs from spin glasses through the Hopfield network and the Boltzmann machine (Edwards and Anderson, 1975; Hopfield, 1982; Amit et al., 1985; Ackley et al., 1985; Smolensky, 1986; Hinton, 2002; Gardner, 1988; Mézard, 2017), a lineage recognised by the 2024 Nobel Prize in Physics (The Royal Swedish Academy of Sciences, 2024); reviews are Zdeborová and Krzakala (2016); Carbone (2025). Its modern branch studies diffusion models with exact scores: for solvable data distributions the reverse dynamics crosses sharp regimes, with a speciation time at which trajectories commit to a mode and, for scores learned from n samples, a memorisation time below which generation collapses onto training points (Biroli and Mézard, 2023; Biroli et al., 2024; Raya and Ambrogioni, 2023; Ghio et al., 2024). On the training side, Bachtis et al. (2024) show that learning in restricted Boltzmann machines proceeds through a cascade of second-order phase transitions: tracking the singular modes of the weight matrix as order parameters, each learned feature corresponds to a Curie–Weiss critical point, the critical times are ordered by the spectrum of the data, and the mean-field signatures (diverging susceptibilities, finite-size scaling, hysteresis) are observed on MNIST, CelebA, and genomic data (building on Saxe et al., 2014; Decelle et al., 2017; Tubiana and Monasson, 2017). Phase transitions thus appear on both the sampling and the training side of generative modelling. What all of these analyses hold fixed is the *internal structure of a single data point*: their coordinates are exchangeable or conditionally i.i.d.; there is no ordering and no dynamics inside one sample.

Inference on chains. Exact posterior inference on Gauss–Markov chains is classical: the Kalman filter and smoother (Kalman, 1960; Rauch et al., 1965), hidden Markov models (Rabiner, 1989), and their unifying formulation as sum–product message passing on factor graphs (Pearl, 1988; Kschischang et al., 2001; Koller and Friedman, 2009; Mézard and Montanari, 2009; Bishop, 2006). This literature fixes the observation noise and asks for the posterior; it does not ask how the posterior, and with it the score (2.8), deforms as a function of the diffusion time t.

Message passing with continuous variables. On trees, belief propagation computes exact marginals; for continuous variables its messages are densities, and practical algorithms restrict them to a parametric family, almost always Gaussian. The classical justification of Gaussian messages, in approximate message passing and the Thouless–Anderson–Palmer equations (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016; Genovese and Piana, 2026), is a central-limit argument valid when every node aggregates many weak incoming messages. On a chain each node has two neighbours, so that argument does not apply, and we are not aware of a quantitative study of the Gaussian message closure on chain-structured diffusion posteriors.

The gap. Sequential data have exactly the structure the solvable-diffusion programme leaves out: an internal time with Markov dependence along it. For such data the score is a smoothing posterior on a chain. How its coupling structure evolves with diffusion time, how it can be computed by message passing, and what a Gaussian message family costs when the data are not Gaussian, are the questions addressed in Chapters 3 and 4.

2.6 How the research questions took shape

The final model and methods were reached through preliminary work whose failures were instructive, and we record the path in compressed form.

Initial question and toy models. The project began by asking how diffusion scores behave for data with an internal time, using small interactive toy models (archived in the project repository): a single distribution transported by the OU flow; a two-frame additive process $x_1 = x_0 + c + \eta$ with a multimodal prior and its two-dimensional joint score; deterministic dynamics (rotations, oscillations) observed through noise. These experiments made the geometry of score fields concrete but exposed a conceptual error: scores were being computed *per frame*, treating a trajectory as K independent samples.

The joint-score correction. Supervision corrected the object of study: a sequence is one sample from a joint law, and the quantity that drives reverse diffusion is the joint score of the whole noisy trajectory. For a stationary chain under the variance-preserving channel the correction is not cosmetic. Each noisy frame is marginally $\mathcal{N}(0, 1)$ at every t, so the per-frame marginal score is $-x_k$, independent of the dynamics; all information about the temporal structure lives in the joint object (Chapter 3 exhibits this explicitly at $K = 2$).

The minimal model. This reframing selected the stationary Gaussian AR(1) chain under coordinatewise OU noise as the minimal model in which the joint score is non-trivial yet exactly computable, with a single coupling parameter α against a single noise parameter t.

The message-passing programme and its bottleneck. Since the score is a smoothing posterior mean (2.8), supervision proposed computing it by belief propagation on the chain's factor graph and asking three questions: whether the recursion reproduces the exact Gaussian score, what happens for non-Gaussian innovations, and how accurate local truncations are. Working out the recursion identified the central obstruction: for continuous variables BP messages are functions, so the number of message updates is linear in K but the cost of an update is unbounded unless the messages close in a finite-dimensional family. This bottleneck fixed the final protocol: prove closure in the Gaussian case, and for Laplace innovations compare a Gaussian-projected recursion against a validated grid-based reference (Chapter 4).

A discarded conjecture. An intermediate conjecture, that the Laplace $K = 2$ posterior Hessian is approximately rank-two, failed under numerical tests outside a narrow regime and was dropped; it is recorded as a negative result in Section 4.6.

Chapter 3

The Gaussian Chain: Exact Score, Belief Propagation, and Locality

This chapter answers RQ1, RQ2, and the Gaussian half of RQ4. For the stationary AR(1) chain under the OU channel we compute the joint score exactly by two independent routes (linear algebra and Bayesian smoothing), characterise how its coupling structure evolves with diffusion time, derive the belief-propagation computation of the same score and identify it with the Kalman filter and smoother, and quantify how rapidly the influence of remote observations decays. Everything in this chapter is exact unless explicitly marked as a numerical check; each boxed formula is covered by a named check of the audit suite (Appendix C).

3.1 The joint law and the object of study

A data point is a trajectory $a = (a_0, \ldots, a_{K-1})$ of the stationary AR(1) chain (2.2) with $\sigma_\eta^2 = 1 - \alpha^2$. Each frame passes independently through the OU channel (2.3) for a time t,

$$x_k \;=\; e^{-t} a_k + \sqrt{\Delta_t}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0,1) \text{ i.i.d.}, \tag{3.1}$$

and the object of study is the joint score of the noisy sequence,

$$S(x,t) \;=\; \nabla_x \log P_t(x) \;\in\; \mathbb{R}^K, \qquad P_t(x) = \int_{\mathbb{R}^K} P_0(a) \prod_{k=0}^{K-1} \mathcal{N}\Big(x_k;\, e^{-t}a_k,\, \Delta_t\Big)\, \mathrm{d}a. \tag{3.2}$$

The gradient is taken with respect to all K noisy frames simultaneously, under the law of the whole noisy trajectory (Section 2.4).

3.2 The clean law: dense covariance, sparse precision

The chain factorisation (2.1) gives the clean joint density

$$P_0(a) = \mathcal{N}(a_0; 0, 1) \prod_{k=0}^{K-2} \mathcal{N}\Big(a_{k+1};\, \alpha a_k,\, \sigma_\eta^2\Big) = \mathcal{N}(a;\, 0,\, \Sigma_0). \tag{3.3}$$

Proposition 3.1 (Stationary covariance). $(\Sigma_0)_{ij} = \alpha^{|i-j|}$*: a dense, symmetric Toeplitz matrix with geometrically decaying off-diagonals (Toolbox 2.1).*

Proposition 3.2 (Tridiagonal precision). $Q_0 := \Sigma_0^{-1}$ *is exactly tridiagonal:*

$$(Q_0)_{kk} = \begin{cases} \dfrac{1}{\sigma_\eta^2}, & k \in \{0, K-1\}, \\[2ex] \dfrac{1+\alpha^2}{\sigma_\eta^2}, & 0 < k < K-1, \end{cases} \qquad (Q_0)_{k,k\pm1} = -\frac{\alpha}{\sigma_\eta^2}, \tag{3.4}$$

and every entry at distance ≥ 2 vanishes identically.

Toolbox 3.1 — Deriving the tridiagonal precision from the log-density

For a centred Gaussian, the precision is the Hessian of the negative log-density. From (3.3),

$$-2\log P_0(a) = \frac{a_0^2}{\sigma_0^2} + \sum_{k=0}^{K-2} \frac{(a_{k+1} - \alpha a_k)^2}{\sigma_\eta^2} + \text{const}, \qquad \sigma_0^2 = 1.$$

Read off the quadratic form term by term. The only term coupling a_k and a_{k+1} is the transition $(a_{k+1} - \alpha a_k)^2/\sigma_\eta^2$, with cross coefficient $-2\alpha/\sigma_\eta^2$, hence precision entry $-\alpha/\sigma_\eta^2$; no term couples variables at distance ≥ 2, so those entries are exactly zero. An interior a_k appears in its left transition with coefficient $1/\sigma_\eta^2$ and in its right transition with α^2/σ_η^2, summing to $(1+\alpha^2)/\sigma_\eta^2$. At the left boundary the prior term $1/\sigma_0^2 = (1-\alpha^2)/\sigma_\eta^2$ plus the right transition α^2/σ_η^2 give $1/\sigma_\eta^2$; the right boundary is symmetric. $\square$

Remark 3.3 (Tridiagonality is Markovianity, not Gaussianity). A zero in the precision matrix is a conditional independence given all other variables; by the Hammersley–Clifford theorem (Besag, 1974), factorisation over the cliques of a graph and conditional independence with respect to that graph are the same property. Gaussianity entered the derivation only to identify the precision with the Hessian of $-\log P_0$; the support of that Hessian, nearest neighbours only, comes from the chain factorisation alone. Any Markov chain with smooth transitions has this sparsity pattern; the Gaussian case makes the entries constants.

The covariance is dense (every frame correlates with every other) while the precision is tridiagonal: correlation is global, conditional dependence is local. The score is governed by the second object.

3.3 The score, two ways

3.3.1 Linear algebra

Proposition 3.4 (Noised law). *$P_t(x) = \mathcal{N}(x;\, 0,\, \Sigma_t)$ with*

$$\boxed{\Sigma_t \;=\; e^{-2t}\,\Sigma_0 \;+\; \Delta_t\, I_K} \tag{3.5}$$

This is (2.4). The diagonal is $e^{-2t}+\Delta_t = 1$ at every t (variance preservation); only the off-diagonal entries, $\alpha^{|i-j|}e^{-2t}$, depend on the diffusion time, because the noise of different frames is independent. Since $\log P_t(x) = -\frac{1}{2}x^\intercal\Sigma_t^{-1}x + \text{const}$,

$$\boxed{S(x,t) \;=\; -\,Q_t\,x, \qquad Q_t := \Sigma_t^{-1}.} \tag{3.6}$$

The joint score is linear in x, the hallmark of Gaussianity, and its entire structure is the noisy precision matrix Q_t: which frames couple, how strongly, and at which t, are questions about this one matrix, answered in Sections 3.4 and 3.7.

3.3.2 Bayesian inference

The score–posterior identity (2.8) expresses the same score through the posterior mean of the clean chain,

$$S_k(x,t) = \frac{e^{-t}\,\mathbb{E}[a_k \mid x] - x_k}{\Delta_t}, \tag{3.7}$$

and the posterior is Gaussian with parameters used throughout:

Proposition 3.5 (Posterior of the clean chain). $P(a \mid x) = \mathcal{N}(a;\, m,\, J^{-1})$ *with*

$$\boxed{J \;=\; \frac{e^{-2t}}{\Delta_t} I \;+\; Q_0, \qquad J\,m \;=\; h := \frac{e^{-t}}{\Delta_t}\,x.} \tag{3.8}$$

Toolbox 3.2 — Posterior precision and field, by completing the square

Bayes' rule with the likelihood (3.1):

$$\begin{aligned} -2\log P(a|x) &= a^{\mathsf{T}} Q_0\, a + \sum_k \frac{(x_k - e^{-t} a_k)^2}{\Delta_t} + \text{const} \\ &= a^{\mathsf{T}}\Big(Q_0 + \tfrac{e^{-2t}}{\Delta_t} I\Big) a - 2\,\tfrac{e^{-t}}{\Delta_t}\, x^{\mathsf{T}} a + \text{const}. \end{aligned}$$

Matching against the generic form $a^{\mathsf{T}} J a - 2h^{\mathsf{T}} a$ gives (3.8). Two structural observations. (i) The likelihood is diagonal in a: observing each frame through its own channel adds the channel precision e^{-2t}/Δ_t to every diagonal entry of the prior precision and touches nothing else. (ii) Consequently J is tridiagonal at every t: the *posterior* of the clean chain remains a nearest-neighbour Markov field; what fills up with t is the precision Q_t of the noisy marginal law. Keeping J (tridiagonal, posterior) and Q_t (dense, noisy marginal) distinct prevents much confusion. □

Substituting $m = J^{-1}h$ into (3.7) must return $-Q_t x$; the operator identity behind this is Woodbury's (Appendix A), and the audit suite checks the two routes agree to 2.7×10^{-15} across sweeps of (K, α, t). The agreement is an end-to-end check of two computations built from different formulas.

3.3.3 The smallest example, $K = 2$

For $K = 2$, with $r := \alpha\, e^{-2t}$,

$$\Sigma_t = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix} \qquad \Longrightarrow \qquad S_0(x,t) = -\frac{x_0 - r\,x_1}{1 - r^2}, \qquad S_1(x,t) = -\frac{x_1 - r\,x_0}{1 - r^2}. \tag{3.9}$$

By contrast, each x_k is marginally $\mathcal{N}(0,1)$ at every t (variance preservation), so the per-frame marginal score is $-x_k$: independent of t and of the other frame, it carries no information about the dynamics. The joint score contains the cross term $-rx_1/(1-r^2)$: the denoiser of frame 0 uses evidence from frame 1 with a weight that equals α at $t = 0$ and decays as e^{-2t}. This is the coupling structure whose full lifecycle the next section derives, in miniature.

3.4 The spectral form and the lifecycle of the precision

How does Q_t interpolate between the tridiagonal Q_0 at $t = 0$ and the identity at $t = \infty$? The analysis rests on two standard facts of linear algebra. Every symmetric $A \in \mathbb{R}^{K\times K}$ has an orthonormal eigenbasis, $A = U\,\mathrm{diag}(\omega_i)U^\intercal$ with $U^\intercal U = I$ (the spectral theorem), so analytic operations act on eigenvalues alone: $A^{-1} = U\,\mathrm{diag}(1/\omega_i)U^\intercal$ when no eigenvalue vanishes. And affine images share the eigenbasis: $(cA + dI)u_i = (c\,\omega_i + d)\,u_i$. Applying the second fact to the forward map (3.5), which is affine in Σ_0, gives the representation from which everything in this section follows.

Theorem 3.6 (Spectral form)**.** *Let $\Sigma_0 = U\,\mathrm{diag}(\omega_i)U^\top$. Then for every $t \ge 0$,*

$$\boxed{\Sigma_t = U\,\mathrm{diag}\left(\lambda_i(t)\right)U^\top, \qquad Q_t = U\,\mathrm{diag}\left(\frac{1}{\lambda_i(t)}\right)U^\top, \qquad \lambda_i(t) = e^{-2t}\omega_i + \Delta_t.} \tag{3.10}$$

Diffusion never rotates the problem: the eigenvectors of Σ_t and Q_t are those of Σ_0, frozen for all t, while each eigenvalue flows along the affine path $\omega_i \mapsto e^{-2t}\omega_i + \Delta_t$, monotonically towards 1. Because Σ_0 is symmetric Toeplitz, its eigenvectors are asymptotically the Fourier modes of the sequence (Brockwell and Davis, 1991): in frequency space the denoising problem is K independent scalar problems at every diffusion time. Two computational corollaries are recorded for later use: factoring Δ_t out of (3.5) gives the SNR form

$$Q_t = \tfrac{1}{\Delta_t}\Big(I + \gamma_t\,\Sigma_0\Big)^{-1}, \qquad \gamma_t := e^{-2t}/\Delta_t, \tag{3.11}$$

and multiplying (3.5) by Q_0 and inverting gives the resolvent form

$$Q_t = e^{2t}\,Q_0\Big(I + c_t Q_0\Big)^{-1} = e^{2t}\sum_{m\ge 0}(-c_t)^m\,Q_0^{m+1}, \qquad c_t := e^{2t} - 1, \tag{3.12}$$

whose Neumann series converges for $\|c_t Q_0\| < 1$, i.e. at small t.

3.4.1 Losing tridiagonality: when, how, and how much

Sparsity is a property of a matrix in a basis. In the eigenbasis the problem is diagonal at all times; in the frame basis, the operationally meaningful one, the zeros of Q_0 are a cancellation that diffusion destroys. The destruction is quantifiable, and it answers three questions: when tridiagonality is lost, how, and how much structure survives at each t.

When: immediately. By (3.10) every entry of Q_t is analytic in t. The distance-d entries, identically zero at $t = 0$ for $d \ge 2$, are non-zero for every $t > 0$: no exact zero survives any positive diffusion time.

How: one diagonal per order in t. The loss is nevertheless graded, and the resolvent form gives the exact rate at which each diagonal switches on.

Theorem 3.7 (Band-fill leading order)**.** *For small t and frame distance $d \geq 1$,*

$$\boxed{(Q_t)_{i,i+d} \;=\; (-1)^{d-1}\,(2t)^{d-1}\left(Q_0^d\right)_{i,i+d} \;+\; O(t^d).} \tag{3.13}$$

Toolbox 3.3 — Band fill from the Neumann series

In the resolvent form (3.12), substitute $c_t = 2t + 2t^2 + O(t^3)$ and $e^{2t} = 1 + O(t)$:

$$Q_t = \Big(1 + O(t)\Big)\Big[Q_0 - c_t Q_0^2 + c_t^2 Q_0^3 - \cdots\Big].$$

Q_0 is tridiagonal, so Q_0^m has bandwidth m: the entry at distance d receives its first contribution from the single term $m = d$, with prefactor $(-c_t)^{d-1} = (-1)^{d-1}(2t)^{d-1}(1+O(t))$; every later term is $O(t^d)$. Each additional frame of coupling range costs one power of t: the band fills outward, one diagonal per order. □

Remark 3.8 (Correction of an earlier formula). An earlier working note of this project stated (3.13) with an extraneous factor $1/(d-1)!$. The factor does not follow from the expansion (the leading distance-d term is a single Neumann term, with no combinatorial sum behind it), and the numerical audit rejects it: the no-factorial coefficient matches the measured $(Q_t)_{i,i+d}/t^{d-1}$ to better than 0.9% for every $d \in \{1, \ldots, 5\}$ (at $\alpha = 0.9$, $K = 25$), while the factorial version fails by 86% already at $d = 3$. The error had survived because slope fits in $\log t$ are blind to constant prefactors; every prefactor in this thesis has since been audited, not only every exponent.

How much: the off-tridiagonal mass. Define the off-tridiagonal Frobenius mass of Q_t: the norm of everything outside the band that Markov structure would allow. At small t the band-fill law makes it rise from zero, dominated by the $d = 2$ diagonal, linearly in t.

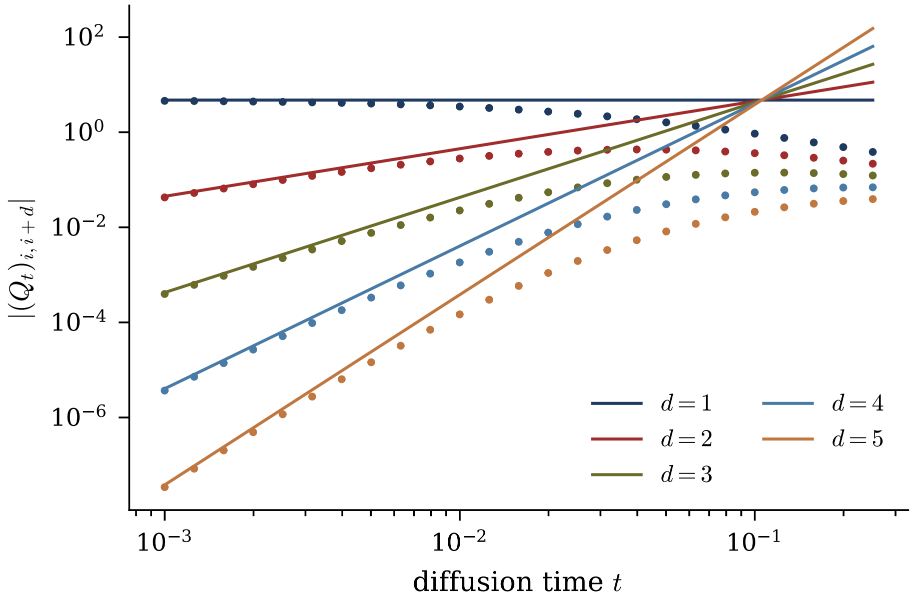

Figure 3.1: Band-fill law. Measured $|(Q_t)_{i,i+d}|$ against t (dots) and the leading order (3.13) (solid lines, no fitted parameters), for $\alpha = 0.9$, $K = 25$, $d = 1, \ldots, 5$. The slopes are $d - 1$; agreement at small t is at the percent level for every d.

At large t, expanding the inverse of $\Sigma_t = I + e^{-2t}(\Sigma_0 - I)$ gives

$$Q_t = I - e^{-2t}(\Sigma_0 - I) + e^{-4t}(\Sigma_0 - I)^2 + O(e^{-6t}), \tag{3.14}$$

so $\|Q_t - I\|_F \simeq e^{-2t}\|\Sigma_0 - I\|_F$: all structure, off-tridiagonal and tridiagonal alike, decays at the rate e^{-2t}, and the score approaches the isotropic $-x$. In between, the off-tridiagonal mass peaks (Figure 3.2): the noisy sequence is most non-Markov at intermediate diffusion times. The lifecycle in one line: tridiagonal at $t = 0$; densifying at rate t^{d-1} per diagonal d; maximally coupled at intermediate t; isotropic as $t \to \infty$ at rate e^{-2t}.

3.5 Belief propagation on the chain

The matrix route costs $O(K^3)$ (or $O(K^2)$ using structure) and is global. The score–posterior identity says the same object is a posterior expectation, and the posterior has a structure that supports local computation. This section develops that computation once,

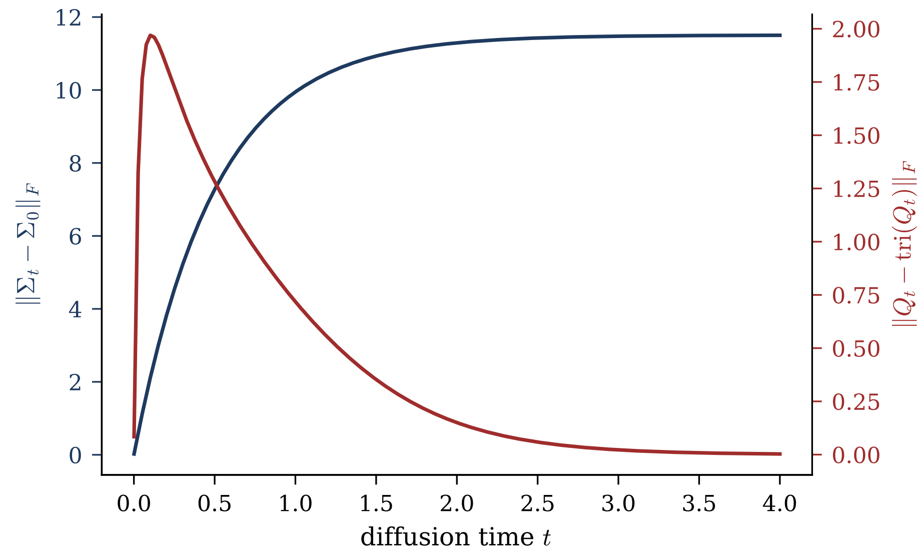

Figure 3.2: Structure across diffusion time ($\alpha = 0.8$, $K = 40$). $\|\Sigma_t - \Sigma_0\|_F$ (navy) grows monotonically as the covariance is driven to the identity; the off-tridiagonal Frobenius mass of Q_t (red) rises from zero, peaks at intermediate t, and decays to zero: the band fills, then empties.

in full.

3.5.1 The factor graph of the posterior

For any Markov-chain prior and per-frame OU evidence, Bayes' rule gives

$$p(a \,|\, x) \;\propto\; \psi_{\mathrm{pr}}(a_0) \prod_{k=0}^{K-2} \psi_{\mathrm{tr}}^{(k)}(a_k, a_{k+1}) \prod_{k=0}^{K-1} \psi_{\mathrm{ob}}^{(k)}(a_k), \qquad \psi_{\mathrm{ob}}^{(k)}(a_k) = \mathcal{N}\Big(x_k;\, e^{-t} a_k,\, \Delta_t\Big), \tag{3.15}$$

a hidden Markov model (Rabiner, 1989): a latent chain observed through independent per-frame noise. By (2.8), the k-th score component requires exactly the smoothing marginal $\mathbb{E}[a_k|x]$ of (3.15).

The representation that makes the factorisation itself graphical is the *factor graph* (Kschischang et al., 2001; Koller and Friedman, 2009): a bipartite graph with a variable node for each a_i, a factor node for each factor ψ, and an edge whenever a variable enters a factor. The factor graph of (3.15) has K variable nodes, $2K$ factor nodes (one prior, $K-1$ transitions, K observations; the data x_k are constants clamped inside their factors) and $3K-1$ edges. A connected graph with one fewer edge than nodes is a tree: here, a spine of variables and transition factors with one evidence leaf per variable.

Proposition 3.9. *For any Markov-chain prior and any per-frame observation model, the posterior $p(a \mid x)$ factorises over a tree-shaped factor graph, and all K smoothing marginals are computed exactly by $O(K)$ message* updates *(one forward and one backward sweep).*

Proposition 3.9 does not say the total *cost* is $O(K)$, because it says nothing about how messages are represented. For discrete chains each message is a probability vector and each update is finite arithmetic; for continuous chains each message is a function, each update is an integral in function space, and the cost per update is unbounded in general. Keeping tree-exactness of the algorithm (a graph property) separate from representability of the messages (a property of the model's function class) is the conceptual hinge of this thesis; the Gaussian case closes the gap, and Chapter 4 measures what happens when it no longer does.

3.5.2 The sum–product recursion, with the bookkeeping explicit

On a factor graph, sum–product belief propagation passes messages along every edge: variable-to-factor messages multiply the incoming factor messages from all other neighbours, and factor-to-variable messages integrate the factor against the incoming variable messages (Pearl, 1988; Kschischang et al., 2001; Mézard and Montanari, 2009). The exclusion of the recipient from each product is the heart of the algorithm: the message from i to a summarises everything i knows except what it learned from a, so no evidence is counted twice along a path. On a tree each edge separates the graph into two halves, the message crossing it summarises exactly the half behind it, and the recursion terminates in one inward and one outward pass with the exact marginals (Mézard and Montanari, 2009, Thm. 14.1). On the chain the schedule collapses into two sweeps, and the bookkeeping deserves care.

Proposition 3.10 (Chain sweeps, Convention A)**.** *Define messages*

$$\mu_{\to 0}(a_0) := \psi_{\mathrm{pr}}(a_0), \quad \mu_{\to k+1}(a_{k+1}) := \int \psi_{\mathrm{tr}}^{(k)}(a_k, a_{k+1})\, \psi_{\mathrm{ob}}^{(k)}(a_k)\, \mu_{\to k}(a_k)\, \mathrm{d}a_k, \tag{3.16}$$

$$\mu_{\leftarrow K-1}(a_{K-1}) := 1, \qquad \mu_{\leftarrow k-1}(a_{k-1}) := \int \psi_{\mathrm{tr}}^{(k-1)}(a_{k-1}, a_k)\, \psi_{\mathrm{ob}}^{(k)}(a_k)\, \mu_{\leftarrow k}(a_k)\, \mathrm{d}a_k. \tag{3.17}$$

Then for every k,

$$p(a_k \,|\, x) \;\propto\; \mu_{\to k}(a_k)\, \psi_{\mathrm{ob}}^{(k)}(a_k)\, \mu_{\leftarrow k}(a_k) : \tag{3.18}$$

the forward message into a_k carries the evidence $x_0, \ldots, x_{k-1}$ only, the backward message carries $x_{k+1}, \ldots, x_{K-1}$ only, and the local evidence enters once, at combination time.

The proof is direct marginalisation: integrating (3.15) over all a_j, $j \neq k$, splits at a_k by the Markov property into independent left and right halves, and peeling the innermost variable of each half yields the recursions.

Remark 3.11 (The double-counting trap)*.* An alternative convention absorbs the local evidence into the forward message, $\mu^{\mathrm{B}}_{\to k} = \mu_{\to k} \cdot \psi_{\mathrm{ob}}^{(k)}$; the product (3.18) then counts x_k twice and must be corrected by dividing one factor back out. An early version of this project's notes fell into this trap, caught by the numerical audit: under Convention A the BP score matches the matrix score to 10^{-14}, under the naive alternative the disagreement is of order one. Every message in this thesis is therefore stated with its evidence content explicit.

3.6 The Gaussian closure and the Kalman smoother

For the Gaussian chain, two elementary facts make the message space finite-dimensional, and they are the only facts needed.

Toolbox 3.4 — The two closure lemmas of Gaussian message passing

Lemma 1 (products). For Gaussians in the same variable,

$$\mathcal{N}(a; m_1, v_1) \times \mathcal{N}(a; m_2, v_2) \;\propto\; \mathcal{N}(a; m, v), \qquad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \qquad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2}:$$

the sum of two quadratics in a is a quadratic in a; precisions and precision-weighted means add.

Lemma 2 (affine marginalisation). Pushing a Gaussian through the AR(1) transition,

$$\int \mathcal{N}(a'; \alpha a, \sigma_\eta^2)\, \mathcal{N}(a; m, v)\, \mathrm{d}a \;=\; \mathcal{N}(a';\, \alpha m,\, \alpha^2 v + \sigma_\eta^2),$$

because $a' = \alpha a + \eta$ is a sum of independent Gaussians. Read in the reverse direction the same computation gives $\mathcal{N}\Big(a;\, m'/\alpha,\, (v' + \sigma_\eta^2)/\alpha^2\Big)$.

The Gaussian family is closed under exactly the two operations the recursions (3.16)–(3.17) perform. The local evidence, read as a function of a_k, is itself Gaussian: $\psi_{\mathrm{ob}}^{(k)}(a_k) \propto \mathcal{N}(a_k;\, e^t x_k,\, \Delta_t e^{2t})$, a pseudo-observation at the deconvolved location $e^t x_k$ with precision e^{-2t}/Δ_t. □

Theorem 3.12 (Gaussian closure of BP on the chain)**.** *For the Gaussian AR(1) chain under the OU channel, every BP message and every belief is exactly Gaussian, at every node, for every* (K, α, t)*. Each message is carried by two numbers, each update is* $O(1)$ *arithmetic, and the two sweeps compute the exact posterior marginals, hence the exact joint score, at total cost* $O(K)$*. Here, and only here, the linear update count of Proposition 3.9 becomes a linear cost.*

Derivation. Induction along each sweep. The initial messages are Gaussian (the prior; the flat message is the zero-precision limit of the family). Each step of (3.16)–(3.17) is one application of Lemma 1 (absorb the local evidence) followed by one of Lemma 2 (push through the transition); the combination (3.18) is Lemma 1 twice. Exactness follows from tree-exactness of sum–product. □

Remark 3.13 (Closure is algebra, not a central limit theorem). The closure holds at node degree two exactly as it would at any degree: it is a property of the Gaussian family under multiplication and affine marginalisation, not an aggregation effect. No central-limit argument is invoked anywhere, so there is nothing for the chain's low degree to break. The central-limit argument matters only for the per-node closures of statistical physics (AMP, the TAP equations), and on the chain it fails in a way that can be made exact; that analysis, complete with a breakdown time $t_c(\alpha)$ and a critical coupling $\alpha_c = \sqrt{2} - 1$, is developed in Appendix B.

Executing the Gaussian sweeps in mean–variance arithmetic reproduces, update by update, the classical algorithms of linear estimation.

Toolbox 3.5 — The Kalman filter, derived from the BP sweep

Write the forward message in Convention A as $\mu_{\to k} = \mathcal{N}(a_k;\, m_{k|k-1},\, P_{k|k-1})$, the predicted state given past observations. One BP step does two things.

Update (absorb the evidence; Lemma 1). Multiplying by the pseudo-observation $\mathcal{N}(a_k; e^t x_k, \Delta_t e^{2t})$ gives the filtered Gaussian $\mathcal{N}(a_k; m_{k|k}, P_{k|k})$ with

$$\frac{1}{P_{k|k}} = \frac{1}{P_{k|k-1}} + \frac{e^{-2t}}{\Delta_t},$$
$$m_{k|k} = P_{k|k}\left(\frac{m_{k|k-1}}{P_{k|k-1}} + \frac{e^{-t}x_k}{\Delta_t}\right) = m_{k|k-1} + G_k\Big(e^t x_k - m_{k|k-1}\Big),$$

where $G_k = P_{k|k}\, e^{-2t}/\Delta_t \in (0,1)$ is the Kalman gain: the filtered mean is the prediction corrected by a gain-weighted innovation (Kalman, 1960).

Predict (push through the dynamics; Lemma 2).

$$m_{k+1|k} = \alpha\, m_{k|k}, \qquad P_{k+1|k} = \alpha^2 P_{k|k} + \sigma_\eta^2.$$

The forward BP sweep is the Kalman filter, message by message.

Smoothing (combine both directions). The belief (3.18) multiplies the filtered forward Gaussian by the backward message (Lemma 1 once more), producing the posterior marginal given all observations, $\mathcal{N}(a_k; m_{k|K-1}, P_{k|K-1})$; carrying out the same algebra for the backward recursion yields the fixed-interval smoother of Rauch et al. (1965). The two-sweep BP schedule on the chain is the Kalman filter followed by the RTS smoother, the Gaussian specialisation of sum–product (Kschischang et al., 2001; Bishop, 2006). □

Combining with the score–posterior identity answers RQ2: the k-th component of the joint diffusion score is an affine function of the Kalman-smoothed estimate of frame k,

$$S_k(x,t) \;=\; \frac{e^{-t}\, m_{k|K-1} - x_k}{\Delta_t}. \tag{3.19}$$

For Markov sequences, score computation is a smoothing problem, and the classical theory of linear smoothing applies directly.

Numerical agreement of the two routes. The audit suite runs BP against the matrix route over 80 configurations ($K \in \{2,3,5,8,16\}$, $\alpha \in \{0.2, 0.5, 0.9, -0.4\}$, $t \in \{0.05, 0.3, 1, 3\}$, random x): maximum disagreement 1.3×10^{-15} on posterior means, 3.9×10^{-15} on variances, 1.2×10^{-14} on scores, i.e. double-precision rounding accumulated along the recursion. An independent implementation, written from scratch against the recursion specification alone, reproduces the same agreement (Appendix C). One qualitative detail from a worked $K=3$ instance: at $x=(1.2,-0.4,0.8)$ the score component S_1 is positive although $x_1 < 0$, because the neighbours pull the posterior mean of the middle frame positive. No per-frame marginal score can produce this; it is inter-frame evidence sharing in numbers.

3.7 The bulk in closed form, and the price of locality

For a long chain the interior (bulk) is translation invariant and can be solved exactly. By Proposition 3.5 the posterior precision in the bulk is tridiagonal Toeplitz with

$$\boxed{J_d \;=\; \frac{e^{-2t}}{\Delta_t} + \frac{1+\alpha^2}{\sigma_\eta^2}, \qquad \beta \;=\; -\frac{\alpha}{\sigma_\eta^2}, \qquad \sigma_\eta^2 = 1-\alpha^2.} \tag{3.20}$$

The diagonal J_d is the sum of the evidence precision e^{-2t}/Δ_t (divergent as $1/2t$ at small t, vanishing at large t) and the constant prior precision; the coupling β is pure prior. The dimensionless ratio $|\beta|/J_d \in (0, \frac{1}{2})$ controls everything below.

Theorem 3.14 (Exact bulk variance and geometric covariance)**.** *In the bulk of a long chain, the posterior variance and covariance are*

$$\boxed{V \;=\; \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \qquad \left(J^{-1}\right)_{i,\,i+d} \;=\; q^{\,d}\, V, \qquad q \;=\; \frac{J_d - \sqrt{J_d^2 - 4\beta^2}}{2\,|\beta|} \in (0,1).} \tag{3.21}$$

Toolbox 3.6 — Transfer-matrix inversion of a tridiagonal Toeplitz matrix

Seek the inverse row in geometric form, $(J^{-1})_{i,i+d} = Cq^{\,d}$ for $d \geq 0$, symmetric in $\pm d$. The defining relation $\sum_j J_{ij}(J^{-1})_{j,i+d} = \delta_{0d}$ reads, for $d \geq 1$,

$$\beta\, Cq^{\,d-1} + J_d\, Cq^{\,d} + \beta\, Cq^{\,d+1} = 0 \quad\Longleftrightarrow\quad \beta q^2 + J_d\, q + \beta = 0,$$

a quadratic whose root inside the unit disc is $q = (J_d - \sqrt{J_d^2 - 4\beta^2})/(2|\beta|)$ (for $\alpha > 0$ all inverse entries are positive; for $\alpha < 0$ they alternate in sign with the same modulus). The $d = 0$ relation fixes the prefactor: $J_d C + 2\beta C q = 1$, and since $2\beta q = -(J_d - \sqrt{J_d^2 - 4\beta^2})$, $C = 1/\sqrt{J_d^2 - 4\beta^2} = V$. Existence of the real square root requires $J_d > 2|\beta|$, which always holds here:

$$J_d - 2|\beta| = \frac{e^{-2t}}{\Delta_t} + \frac{1+\alpha^2 - 2|\alpha|}{\sigma_\eta^2} = \frac{e^{-2t}}{\Delta_t} + \frac{(1-|\alpha|)^2}{\sigma_\eta^2} \;>\; 0$$

for every $|\alpha| < 1$ and $t > 0$. This positivity margin is also what guarantees the belief-propagation fixed point exists for every coupling and diffusion time (Appendix B). $\square$

The number $q(\alpha, t)$ measures how far evidence travels along the chain: the posterior correlation length is $\xi = 1/\log(1/q)$. Its limits are

$$\begin{aligned} t \to 0: &\quad q \approx \frac{|\beta|}{J_d} \to 0 \quad \text{(each frame pinned by its own observation);} \\ t \to \infty: &\quad q \to |\alpha| \qquad\quad \text{(posterior} \to \text{prior),} \end{aligned} \tag{3.22}$$

the second because $q \to \big((1+\alpha^2)-(1-\alpha^2)\big)/(2|\alpha|) = |\alpha|$ when the evidence term vanishes. As diffusion time runs, the posterior correlation length climbs monotonically from 0 to the prior's own $1/\log(1/|\alpha|)$.

3.7.1 The locality law

RQ4 asks how rapidly the influence of remote observations decays. Define the local estimator of radius r: the posterior mean of a_k given only the observations within distance r, $m_k^{(r)} = \mathbb{E}[a_k \mid x_{k-r}, \ldots, x_{k+r}]$. Two facts make the comparison clean. Any contiguous window of a stationary AR(1) chain is itself a stationary AR(1) chain, so $m_k^{(r)}$ is exact on its window; the only approximation is the truncation of range. And both estimators are linear in x, so the error $m_k - m_k^{(r)} = e^\top x$ has a closed-form RMS over the data law, $\mathrm{RMS}_r = \sqrt{e^\top \Sigma_t e}$, a deterministic number with no sampling involved.

Theorem 3.15 (Locality error decays as q^r)**.** *In the bulk,* $\log \mathrm{RMS}_r$ *decreases linearly in* r *with slope* $\log q(\alpha, t)$*: truncating the evidence range costs a factor* q *per discarded frame.*

The mechanism is Theorem 3.14: the influence of evidence at distance d on the posterior mean is proportional to $(J^{-1})_{k,k\pm d} \propto q^d$, and the leading discarded evidence sits at distance $r+1$. The audit verifies the rate: fitted slopes of $\log \mathrm{RMS}_r$ over $r = 2, \ldots, 13$ at the centre of a $K = 121$ chain match $\log q$ within 0.2% at $t \in \{0.1, 0.5, 2.0\}$ ($\alpha = 0.8$); see Figure 3.3.

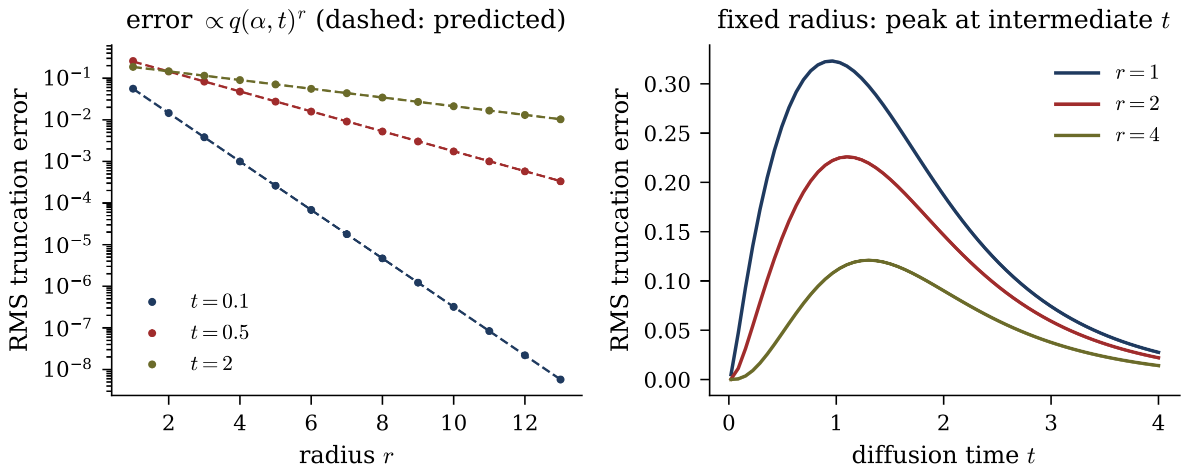

Figure 3.3: Locality of the joint score ($\alpha = 0.8$, $K = 121$, deterministic RMS over the data law). Left: exact truncation error of the radius-r local estimator (dots) at three diffusion times; the dashed lines have the predicted slopes $q(\alpha, t)^r$ from (3.21), with no fitting. Right: the same error at fixed radius as a function of t: it vanishes at both ends and peaks at intermediate t, where the band of Q_t is fullest.

3.7.2 Truncating the matrix versus truncating the inference

A truncated score can be built in two inequivalent ways. *Truncate the answer*: band the matrix, $\widehat{S} = -Q_t^{(b)}x$ with entries beyond $|i-j| \le b$ zeroed. *Truncate the algorithm*: let messages travel at most r hops, i.e. run exact inference on the window $x_{k-r}, \dots, x_{k+r}$ (the local estimator above). Both are linear in x, so their RMS errors are again deterministic. At equal locality budget ($b = r = 1$, $K = 40$, $\alpha = 0.8$):

relative RMS score error	$t = 0.05$	$t = 0.3$	$t = 1$	$t = 3$
banded matrix ($b = 1$)	0.210	0.302	0.135	0.0035
truncated inference ($r = 1$)	0.123	0.235	0.130	0.0035

Two observations. The cost of locality is non-monotone in t and peaks at intermediate times, consistent with Section 3.4.1. And at small t truncating the algorithm beats truncating the answer (12.3% against 21.0% at $t = 0.05$): the windowed inference solves a correctly normalised problem on its window, while zeroing entries of an inverse produces a matrix that is the inverse of nothing and misstates the diagonal. The two coincide at large t, where the off-diagonal content vanishes. Either error decays geometrically in the allowed range at the rate $q(\alpha, t)$ of Theorem 3.15.

3.8 Status of the results

Exact, by derivation: the score formulas (3.6) and (3.7), the spectral form (3.10), the band-fill law (3.13) as a small-t asymptotic, the large-t law (3.14), tree exactness and Gaussian closure of BP, the Kalman identification, and the bulk formulas (3.20)–(3.21) with the locality rate. Numerical, within stated grids and tolerances: the agreement of the two score routes, the audited band-fill coefficients, the locality slopes, and the truncation table. Interpretation, limited to this model: within the Gaussian benchmark, a local score computation with receptive field $r \gtrsim \xi \log(1/\varepsilon)$, $\xi = 1/\log(1/q)$, achieves relative error ε, and the requirement is most demanding at intermediate diffusion times; whether trained score networks on real sequence data behave accordingly is not tested here.

Chapter 4

Beyond Gaussianity: Laplace Innovations and the Gaussian-Message Approximation

Every exact result of Chapter 3 rests on the closure of the Gaussian family under the two operations the model performs, multiplying densities and marginalising. This chapter removes that property in the most controlled way available and answers RQ3. We keep the AR(1) recursion, the OU channel, and the factor-graph structure, and change only the innovation law from Gaussian to Laplace: heavy-tailed and kinked at the origin, yet log-concave and analytically tractable, it is a minimal step outside the Gaussian family, so every deviation we find is attributable to non-Gaussianity alone. The chapter first establishes what remains exact (the score–posterior identity, the tree structure, one boundary integral, and a differential characterisation of the $K = 1$ marginal), then locates precisely where closed forms stop, and finally measures, against a validated numerical reference, what a Gaussian message family costs.

4.1 What changes, and what survives

The Laplace AR(1) chain replaces the Gaussian innovations of (2.2) with Laplace ones:

$$P_0(a) = \frac{1}{(2b)^K} \exp\!\left(-\frac{1}{b}\sum_{k=0}^{K-1} \Big|a_k - \alpha a_{k-1}\Big|\right), \qquad a_{-1} := 0, \tag{4.1}$$

observed through the same per-frame OU channel. Three structural facts survive unchanged, because none of them used Gaussianity of the prior. The score–posterior identity (2.8) holds for any p_0. The posterior factorises over the same tree-shaped factor graph (Proposition 3.9), so the sum–product recursion (3.16)–(3.17) still computes exact smoothing marginals in $O(K)$ message updates. And the OU likelihood factors are still Gaussian in a_k. What is lost is representability: the messages are no longer Gaussian, no finite-dimensional family we know of is closed under the updates, and the cost of an update is no longer bounded. The score also ceases to be linear in x, which is the first thing to make precise.

4.2 One frame: the score becomes nonlinear

For $K = 1$ there is no chain; the model isolates the ingredient "non-Gaussian prior meets Gaussian channel". Let $a \sim \mathrm{Laplace}(0, b)$, $p_0(a) = e^{-|a|/b}/2b$, and write $\mu := e^{-t}$. The noised marginal is the convolution

$$p_t^{(1)}(x) = \int p_0(a)\, \mathcal{N}\Big(x;\, \mu a,\, \Delta_t\Big)\, \mathrm{d}a. \tag{4.2}$$

As $t \to 0$ the Gaussian kernel concentrates and $p_t^{(1)} \to p_0$; as $t \to \infty$, $p_t^{(1)} \to \mathcal{N}(0, 1)$, the OU attractor. By the score–posterior identity, $S(x, t) = (\mu\, \mathbb{E}[a|x] - x)/\Delta_t$; for a Laplace prior the conditional mean is a nonlinear function of x, and the score inherits the

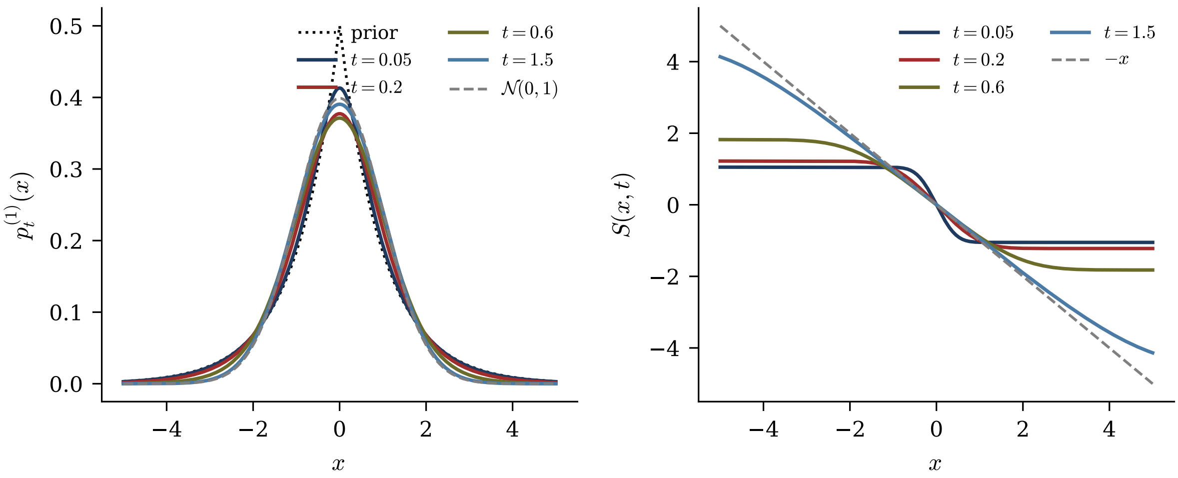

Figure 4.1: The Laplace $K = 1$ problem ($b = 1$). Left: noised marginal $p_t^{(1)}$ at increasing diffusion times, from the kinked Laplace prior (dotted) to the Gaussian attractor (dashed). Right: the score at the same times; the dashed line is the attractor $-x$. The nonlinearity lives near $x = 0$ and flattens as t grows; the limits are the exact statements (4.3). Curves computed by deterministic quadrature.

nonlinearity. Its limits are exact: for every fixed $x \neq 0$,

$$t \to 0: \quad S(x,t) \longrightarrow \partial_x \log p_0(x) = -\frac{\operatorname{sign}(x)}{b}, \qquad t \to \infty: \quad S(x,t) \longrightarrow -x. \tag{4.3}$$

At small t the score converges pointwise to the sign-shaped score of the prior, with the nonlinearity concentrated in the kink near $x = 0$; at large t it flattens into the linear attractor score $-x$ (Figure 4.1). The audit verifies the posterior-expectation route against central finite differences of $\log p_t^{(1)}$ computed by quadrature: maximum disagreement 1.19×10^{-7} over a grid of (b, t, x), at the truncation floor of the finite difference.

4.2.1 A differential characterisation of the noised marginal

The channel writes x as a sum of independent variables, so (4.2) is a convolution, which factorises in Fourier space:

$$\widehat{p}_t^{(1)}(k) = \frac{1}{1 + b^2\mu^2k^2}\, e^{-\Delta_t k^2/2}, \tag{4.4}$$

the product of the (rescaled) Laplace and Gaussian characteristic functions.

Theorem 4.1 (Screened Poisson equation)**.** *The Laplace noised marginal solves the linear, constant-coefficient ODE*

$$\boxed{\left(1 - b^2\mu^2\,\partial_x^2\right) p_t^{(1)}(x) \;=\; G_{\Delta_t}(x),} \qquad G_{\Delta_t}(x) = \frac{e^{-x^2/2\Delta_t}}{\sqrt{2\pi\Delta_t}}. \tag{4.5}$$

Toolbox 4.1 — From the characteristic function to the screened Poisson equation

Multiply (4.4) by $(1 + b^2\mu^2k^2)$:

$$\left(1 + b^2\mu^2k^2\right)\widehat{p}_t^{(1)}(k) = e^{-\Delta_t k^2/2},$$

whose right-hand side is the characteristic function of G_{Δ_t}. Multiplication by k^2 in Fourier space is $-\partial_x^2$ in real space, so the polynomial $(1+b^2\mu^2k^2)$ becomes the operator $(1-b^2\mu^2\partial_x^2)$, and (4.5) follows by inverse transformation. The audit checks (4.4) against direct quadrature of the characteristic function to 2.9×10^{-6}, the floor of oscillatory quadrature. □

Equation (4.5) states that the noised marginal is the solution of a screened Poisson equation with a Gaussian source, with screening length $b\mu = be^{-t}$: the memory of the prior's tails shrinks as e^{-t}. The deviation from Gaussianity is governed by a single local differential operator whose strength dies with diffusion time.

One further exact consequence identifies what a Gaussian approximation must get wrong. Differentiating the score–posterior identity once more,

$$H_t(x) := -\partial_x^2 \log p_t^{(1)}(x) = \frac{1}{\Delta_t}\left(1 - \frac{\mu^2}{\Delta_t}\,\mathrm{Var}[a\,|\,x]\right). \tag{4.6}$$

For a Gaussian prior the posterior variance is independent of x and H_t is constant. For the Laplace prior $\mathrm{Var}[a|x]$ depends on x and H_t is a nontrivial function: the breakdown of a constant curvature is the second-order form of the breakdown of score linearity. A Gaussian message is exactly the object that cannot represent an x-dependent curvature, so (4.6) states what the Gaussian closure of Section 4.5 will miss, before it is measured.

4.3 The chain: where closed forms stop

Theorem 4.2 (Joint score from chain messages)**.** *For any chain prior, in particular* (4.1)*, and the OU likelihood, the joint score is, componentwise,*

$$S_k(x,t) = \frac{\mu\,\mathbb{E}[a_k|x] - x_k}{\Delta_t}, \qquad \mathbb{E}[a_k|x] = \frac{\int a_k\ \mu_{\to k}(a_k)\ \psi^{(k)}_{\mathrm{ob}}(a_k)\ \mu_{\leftarrow k}(a_k)\ \mathrm{d}a_k}{\int \mu_{\to k}(a_k)\ \psi^{(k)}_{\mathrm{ob}}(a_k)\ \mu_{\leftarrow k}(a_k)\ \mathrm{d}a_k}, \tag{4.7}$$

with the messages of Proposition 3.10.

This is a complexity statement: the K-dimensional posterior integral is replaced by two one-dimensional message recursions plus one one-dimensional integral per frame. Computational locality survives non-Gaussianity; what does not survive is the collapse of those one-dimensional integrals into closed-form arithmetic. Exactly one such integral has a closed form.

Proposition 4.3 (One-step boundary integral)**.** *For the Laplace transition* $M(a'|a) = \frac{1}{2b}e^{-|a'-\alpha a|/b}$ *and OU evidence* $\mathcal{N}(x';\mu a',\Delta_t)$*,*

$$\int_{\mathbb{R}} M(a'|a)\ \mathcal{N}(x';\mu a',\Delta_t)\ \mathrm{d}a' \ = \ p_t^{(1)}\Big(x' - \mu\alpha\, a\Big) : \tag{4.8}$$

the first backward message of the chain is the $K = 1$ *noised marginal evaluated at a shifted argument.*

The proof is the innovation substitution $r := a' - \alpha a$, which turns the transition into the bare Laplace density at r and the evidence into $\mathcal{N}(x' - \mu\alpha a;\, \mu r,\, \Delta_t)$; the integral is then the definition (4.2). The shift $\mu\alpha a$ is what frame a predicts for the next noisy observation, and the whole $K = 1$ analysis, screened Poisson equation included, recycles as the chain's boundary building block.

Remark 4.4 (Why iteration destroys the closed form). Equation (4.8) works because the integrand contains exactly one Laplace kink times exactly one Gaussian. Every subsequent BP step violates this: the incoming message already carries the propagated structure of several frames, so the next integrand contains two or more absolute-value kinks. On each sign-quadrant the integral reduces to half-line Gaussians, but globally it is a sum of error functions whose arguments depend on the conditioning variable non-affinely, and the clean form does not survive one iteration. In the Gaussian chain every step is Gaussian-times-Gaussian and closure propagates forever; this is the precise structural location of the difference between the two models, and the reason the Gaussian projection below is an approximation with an error to be measured, not an identity.

A change of coordinates shows the obstruction is intrinsic, not an artefact of the frame basis. In innovation coordinates $r_0 = a_0$, $r_k = a_k - \alpha a_{k-1}$ (unit Jacobian), the prior factorises into i.i.d. Laplace variables, but the likelihood becomes $p(x|r) \propto \exp(-\frac{\mu^2}{2\Delta_t} r^\top L^\top L r + \frac{\mu}{\Delta_t} x^\top L r)$ with L the unit lower-triangular map $a = Lr$, and $L^\top L$ is dense. The coupling does not disappear; it migrates from the prior to the likelihood. There is no basis in which both are diagonal at once.

4.4 A validated numerical reference: grid BP

Since functional messages admit no closed form, the exact recursion is approximated numerically: messages are represented by their values on M equispaced points in $[-A, A]$ and the update integrals are carried out by trapezoidal quadrature, at cost $O(KM^2)$ per sweep. This *grid BP* is a numerical approximation of the functional-message recursion, and it can serve as a reference only inside a validated regime, which we establish on the exactly solvable Gaussian chain. There the trapezoid rule on analytic, rapidly decaying integrands converges spectrally, so failure is abrupt rather than gradual: at $M = 101$, $A = 8$ the score error sits at the 10^{-15} floor down to $t = 0.02$ and fails only when the likelihood width $\sqrt{2t}$ approaches the grid step. The working configuration for all non-Gaussian experiments is $M = 401$, $A = 8$, with a reference solution at $M = 801$ and the resolution constraint $\sqrt{2t} \gtrsim 3\,\mathrm{d}x$ checked automatically; grid self-convergence error remains 10^4 to 10^7 times smaller than every effect measured below. Within this regime we treat grid BP as the reference score; outside it no claim is made.

All score errors are amplified posterior-mean errors: for any two solvers with means $\widehat{m}$ and m_{ref}, the score–posterior identity gives, exactly,

$$\widehat{s} - s_{\mathrm{ref}} = \frac{e^{-t}}{\Delta_t}\Big(\widehat{m} - m_{\mathrm{ref}}\Big), \qquad \frac{e^{-t}}{\Delta_t} \sim \frac{1}{2t} \ (t \downarrow 0). \tag{4.9}$$

The numerical residual of this identity is logged in every experiment and stays at 10^{-15} to 10^{-12}. Equation (4.9) explains in advance why small diffusion times are the difficult regime: whatever an approximation does to the posterior mean is amplified by $1/2t$ in the score.

4.5 The Gaussian-message approximation, measured

The object under test is Gaussian-projected BP: the same sweeps, with every message forced into the Gaussian family by moment matching. This is the algorithm one would deploy when quadrature is unaffordable, and the chain analogue of the per-node closures of statistical physics (Appendix B).

On the Laplace chain (chain correlation $\alpha = 0.85$, $K = 40$ frames, 60 trials per t, reference $M = 801$), the median relative score error of the Gaussian closure is

t	0.02	0.08	0.4	2.4
median relative score error	0.39	0.17	3.5×10^{-2}	7.6×10^{-5}

with failure probability $\mathbb{P}(\mathrm{err} > 0.1)$ equal to 1 for $t \leq 0.08$ and 0 for $t \geq 0.9$ (Figure 4.2). The closure error is of order tens of percent at small diffusion time and decays below 10^{-3} once diffusion noise Gaussianises the tilted posteriors. Inspection of the worst trials shows the mechanism: the reference belief is sharply non-Gaussian, occasionally bimodal; the single Gaussian commits to a compromise mean; and (4.9) amplifies the shift. The score *direction* is much more robust than its magnitude: the median cosine similarity between

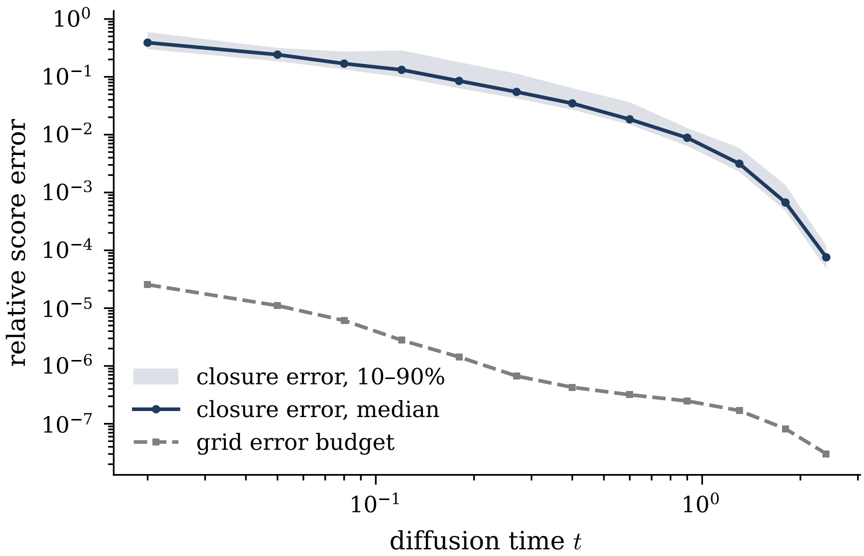

Figure 4.2: Gaussian-closure score error on the Laplace chain ($\alpha = 0.85$, $K = 40$, 60 trials per t; median and 10–90% band) against the grid error budget measured on identical data. The closure error exceeds the reference error by 10^4 to 10^7 at every t, so the measurement is trustworthy where it is used.

projected and reference scores is 0.92 at $t = 0.02$ (where the magnitude error is 39%), exceeds 0.99 for all $t \geq 0.12$, and the worst single trial of the sweep stays above 0.78. These are numerical observations for the stated parameter ranges, not theorems.

4.5.1 Varying the innovation law

Sweeping six innovation families with identical covariance $\alpha^{|i-j|}$ (Laplace, Student-t with $\nu = 5, 8$, and symmetric Gaussian mixtures of increasing separation) isolates the shape of the innovation law as the only moving part. Three observations, all within the tested grids. First, the noise level dominates: every family's error decays to about 10^{-3} or below by $t = 2.4$; all substantial closure error lives at $t \lesssim 0.4$. Second, bimodality is the worst case: at $t = 0.05$, $\alpha = 0.5$, the median error is 0.94 for a strongly bimodal mixture, 0.52 for Laplace, 0.32 for Student-$t_{\nu=5}$, and 0.07 for a mildly bimodal mixture; a single Gaussian cannot represent two posterior modes and commits to the gap between them. Third, stronger chain correlation helps rather than hurts: for Laplace at $t = 0.05$ the error is 0.43, 0.27, 0.13 at $\alpha = 0.5$, 0.85, 0.95; informative neighbours act like additional

Gaussian observations and Gaussianise the local tilted posteriors.

4.6 What can and cannot be concluded

Exact: the surviving structure of Section 4.1, the limits (4.3), the factorisation (4.4), the screened Poisson equation (4.5), the curvature identity (4.6), the message form (4.7), the boundary integral (4.8), the breakdown argument of Remark 4.4, and the error identity (4.9). Numerical, within validated regimes: the quadrature audits of Section 4.2, the grid-BP calibration, and the closure-error measurements, which are specific to the stated innovation laws, correlation levels, chain lengths, and diffusion times. Not concluded: closed forms for general (non-boundary) messages at $K \geq 2$; any locality law for the non-Gaussian score beyond the structural delimitation above; and any statement about learned scores. An earlier conjecture of this project, that the Laplace $K = 2$ posterior Hessian is approximately rank-two, proved regime-dependent under numerical tests and is not asserted.

Chapter 5

Discussion and Conclusions

5.1 Findings

Derived exactly. For the stationary AR(1) chain under coordinatewise OU corruption, the joint score is $S(x,t) = -Q_t x$ with $Q_t = (e^{-2t}\Sigma_0 + \Delta_t I)^{-1}$, equal by the score–posterior identity to $(e^{-t}\mathbb{E}[a|x] - x)/\Delta_t$ (RQ1). Its coupling structure has a closed-form lifecycle: exactly tridiagonal at $t = 0$, filling one diagonal per order in t by the band-fill law (3.13), and returning to the identity at rate e^{-2t}. In the bulk, the single quantity $q(\alpha,t) \in (0, |\alpha|]$ governs both the posterior correlation length and the geometric decay of evidence influence, with limits $q \to 0$ as $t \to 0$ and $q \to |\alpha|$ as $t \to \infty$. The posterior of the noisy chain lives on a tree-shaped factor graph, on which the two BP sweeps compute all smoothing marginals in $O(K)$ message updates; on the Gaussian chain the messages close algebraically in the Gaussian family, the recursion is the Kalman filter followed by the RTS smoother, and the cost becomes $O(K)$ (RQ2). The error of a radius-r local score estimator decays exactly as q^r (RQ4). On the same model, the per-node AMP/TAP closure returns the exact score (the mean fixed point is closure-independent) but a different variance, with existence boundary, breakdown time $t_c(\alpha)$, and critical coupling $\alpha_c = \sqrt{2} - 1$ in closed form (Appendix B); all of these formulas were re-verified independently.

Measured numerically. The two Gaussian score routes agree to 10^{-15}–10^{-14} across the tested (K, α, t) grid, with an independently written implementation reproducing the agreement. On Laplace chains, Gaussian-projected BP measured against validated grid quadrature has median relative score error 0.39 at $t = 0.02$, decaying below 10^{-3} for $t \gtrsim 0.9$; the error is amplified at small t by the exact factor e^{-t}/Δ_t, is worst for bimodal innovation laws, and decreases with stronger chain correlation, while the score direction remains accurate (median cosine ≥ 0.92 throughout) (RQ3). At equal locality budget, truncating the inference is more accurate than truncating the score matrix at small t (12.3% against 21.0% at $t = 0.05$).

The Gaussian-to-Laplace comparison. The structural lesson is where exactness is lost: the tree, the two-sweep schedule, and the score–posterior identity survive any innovation law; what breaks is the representability of messages in a finite-dimensional family. The practical lesson is when it matters: the Gaussian closure is essentially free at moderate and large diffusion times, and expensive precisely in the small-t regime where the score magnitude is amplified, though even there the score direction is preserved much better than its magnitude.

5.2 What the model suggests but does not establish

Within the Gaussian benchmark, local score computations are near-optimal away from intermediate diffusion times, the required receptive field scales as $\xi \log(1/\varepsilon)$ with $\xi = 1/\log(1/q)$, and a local algorithm is preferable to a banded linear readout. These are properties of exact inference in one controlled model. Whether trained score networks on real sequence data inherit them is untested here and would require learning experiments.

5.3 Limitations

(i) The exact solutions concern one-dimensional state per frame and linear AR(1) dynamics. (ii) The non-Gaussian results are numerical: grid BP is a reference only within its validated resolution regime ($\sqrt{2t} \gtrsim 3\,\mathrm{d}x$), and the Laplace closure error is a measured curve, not a formula. (iii) The locality laws are proved for the Gaussian bulk; their non-Gaussian counterparts are only structurally delimited. (iv) No learning is performed. (v) An earlier conjecture on low-rank structure of the Laplace $K = 2$ posterior Hessian failed outside a narrow regime and is not asserted. (vi) How score errors accumulate along the reverse generation trajectory was not studied.

5.4 Future work

Three directions follow directly from what was measured. First, mixture messages: the single-Gaussian closure fails worst on bimodal beliefs, so a two-component mixture closure on the same sweeps would separate the representation error from the model mismatch, and is the clearest next algorithmic step. Second, discrete chains: with a finite alphabet the BP messages are finite vectors and inference is exact with no closure at all, giving a second solvable benchmark. Third, reverse dynamics: integrating the reverse SDE under exact, Gaussian-projected, and truncated scores would show where along the trajectory the closure and locality errors matter for generation, turning the pointwise error measurements of this thesis into dynamical ones.

Appendix A

Gaussian Identities Used Throughout

This appendix collects, with proofs, the handful of Gaussian identities on which Chapter 3 relies. Notation: a Gaussian density in *moment* parameters is $\mathcal{N}(a; m, v)$; in *natural* parameters it is $\exp(-\frac{\lambda}{2}a^2 + \theta a + \text{const})$ with $\lambda = 1/v$, $\theta = m/v$.

A.1 Quadratic forms and Gaussian normalisation

For $J \succ 0$ (symmetric positive definite) and any h,

$$\int_{\mathbb{R}^K} \exp\Big(-\tfrac{1}{2}a^\mathsf{T} J a + h^\mathsf{T} a\Big)\,\mathrm{d}a = \frac{(2\pi)^{K/2}}{\sqrt{\det J}} \exp\Big(\tfrac{1}{2}h^\mathsf{T} J^{-1} h\Big), \tag{A.1}$$

by completing the square, $-\frac{1}{2}a^\mathsf{T} J a + h^\mathsf{T} a = -\frac{1}{2}(a - J^{-1}h)^\mathsf{T} J (a - J^{-1}h) + \frac{1}{2}h^\mathsf{T} J^{-1} h$, and translating. Consequently a density $p(a) \propto \exp(-\frac{1}{2}a^\mathsf{T} J a + h^\mathsf{T} a)$ is the Gaussian with mean $m = J^{-1}h$ and covariance J^{-1}: *reading off* (J, h) *from a log-density is the fastest route to posterior parameters*, used in Proposition 3.5.

A.2 Products of Gaussians in the same variable

$$\mathcal{N}(a; m_1, v_1)\,\mathcal{N}(a; m_2, v_2) \;\propto\; \mathcal{N}\Big(a;\, m, v\Big), \qquad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \qquad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2}. \quad \text{(A.2)}$$

Proof. Add the exponents; the sum of two quadratics in a is a quadratic in a whose a^2-coefficient is $-\frac{1}{2}(1/v_1 + 1/v_2)$ and whose linear coefficient is $m_1/v_1 + m_2/v_2$; apply (A.1) in one dimension. In natural parameters the rule is addition: $(\lambda, \theta) = (\lambda_1 + \lambda_2, \theta_1 + \theta_2)$, which is why the BP updates of Chapter 3 are two additions per message. □

A.3 Affine propagation (marginalisation)

If $a' = \alpha a + \eta$ with $\eta \sim \mathcal{N}(0, \sigma^2)$ independent of $a \sim \mathcal{N}(m, v)$, then

$$\int \mathcal{N}(a'; \alpha a, \sigma^2)\,\mathcal{N}(a; m, v)\,\mathrm{d}a = \mathcal{N}\Big(a';\, \alpha m,\, \alpha^2 v + \sigma^2\Big). \tag{A.3}$$

Proof. a' is a sum of independent Gaussians, hence Gaussian; its mean is αm and its variance $\alpha^2 v + \sigma^2$ by linearity of expectation and independence. Read in the reverse direction (solving for a given a Gaussian summary of a'), the same computation gives $\mathcal{N}\Big(a;\, m'/\alpha,\, (v' + \sigma^2)/\alpha^2\Big)$, the backward half-step of the BP sweep. □

A.4 Gaussian conditioning

For a jointly Gaussian pair with precision blocks $\left(\begin{smallmatrix} J_{11} & J_{12} \\ J_{12}^{\top} & J_{22} \end{smallmatrix}\right)$, the conditional of block 1 given block 2 is Gaussian with precision J_{11} (conditioning *reads off* the precision block) and mean shifted by $-J_{11}^{-1} J_{12}\, a_2$; marginalisation, dually, is trivial in covariance parameters. This precision/covariance duality (conditioning is local in J, marginalising is local in Σ) is the algebraic engine behind the cavity recursion (B.1): integrating a neighbour out of a tridiagonal precision produces the rank-one correction $-\beta^2/\lambda$ on the adjacent diagonal entry, by the Schur complement

$$\begin{pmatrix} \lambda & \beta \\ \beta & J_d \end{pmatrix} \longrightarrow J_d - \frac{\beta^2}{\lambda}. \tag{A.4}$$

A.5 Woodbury and the two routes to the score

The Woodbury identity $(A+UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1}+VA^{-1}U)^{-1}VA^{-1}$, specialised to $\Sigma_t = \Delta_t I + e^{-2t}\Sigma_0$, connects the noisy precision Q_t and the posterior precision $J = (e^{-2t}/\Delta_t)I + Q_0$:

$$Q_t = \frac{1}{\Delta_t}\left(I - \frac{e^{-2t}}{\Delta_t}J^{-1}\right), \tag{A.5}$$

which is exactly the operator form of the score–posterior identity: substituting the posterior mean $m = J^{-1}h = J^{-1}(e^{-t}/\Delta_t)x$ into $S = (e^{-t}m - x)/\Delta_t$ gives $S = -Q_t x$ identically. The machine-precision agreement of the two score routes (Section 3.3) is the numerical shadow of (A.5).

Appendix B

AMP, TAP, and the Bulk Fixed Point

This appendix develops the per-node Gaussian closures of statistical physics, AMP and the TAP equations, on the Gaussian chain, and proves the exact breakdown announced in Remark 3.13. The results are complete, audited, and independently re-verified; they are collected here rather than in Chapter 3 because they branch off the thesis's main line. Summary: on the chain, AMP still returns the exact score, but its variance closure fails past an exact breakdown time $t_c(\alpha)$, with critical coupling $\alpha_c = \sqrt{2} - 1$.

B.1 The bulk cavity fixed point of BP

Chapter 3 solved the bulk covariance by matrix algebra; the same formulas re-emerge from the message recursion, which is worth exhibiting because it localises where BP and AMP differ. In the bulk, parametrise the forward message (evidence absorbed) by its precision λ_k; the closure lemmas of Section 3.6 give the scalar recursion

$$\lambda_{k+1} \;=\; J_d - \frac{\beta^2}{\lambda_k}, \tag{B.1}$$

with J_d, β the bulk parameters (3.20): absorbing one neighbour's summary costs the Gaussian-integration correction $-\beta^2/\lambda$. The fixed points are $\lambda^\pm = \left(J_d \pm \sqrt{J_d^2 - 4\beta^2}\right)/2$, and since $J_d - 2|\beta| = e^{-2t}/\Delta_t + (1-|\alpha|)^2/\sigma_\eta^2 > 0$ (Chapter 3), the attractive upper fixed point

$$\boxed{\lambda^* = \frac{J_d + \sqrt{J_d^2 - 4\beta^2}}{2}} \tag{B.2}$$

exists for every $\alpha \in (-1, 1)$ and $t > 0$, with $|f'(\lambda^*)| = \beta^2/\lambda^{*2} < 1$: the sweep converges geometrically from any initialisation. Recombining the two directions and the local term, and using the fixed-point relation $\beta^2/\lambda^* = J_d - \lambda^*$,

$$\Lambda_{\text{bulk}} = J_d - \frac{2\beta^2}{\lambda^*} = 2\lambda^* - J_d = \sqrt{J_d^2 - 4\beta^2} \quad \Longrightarrow \quad V_{\text{exact}} = \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \tag{B.3}$$

in exact agreement with Theorem 3.14. Belief propagation on the Gaussian chain has no phase transition: the cavity fixed point exists always, and the discriminant $J_d^2 - 4\beta^2$ is the quantity to watch as the closure is modified.

B.2 AMP/TAP: same score, different variance

AMP, the algorithmic form of the TAP equations (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016), is the message passing one uses when the factor graph is dense: each message is then a sum of many weak contributions, approximately Gaussian by a central-limit argument, and one tracks per-node means and variances instead of per-edge messages, with the Onsager term correcting the self-influence. Rigorous derivations of AMP from BP exist in that dense limit (Genovese and Piana, 2026). The chain is the opposite regime in every respect (diagonal measurement, coupling in the prior, degree two), so it isolates what the per-node closure costs. Viewing the posterior as a Gaussian Markov field with precision J and field h, the three schemes solve, at each node,

$$\text{mean field: } V_i = \frac{1}{J_{ii}}; \qquad \text{AMP/TAP: } V_i = \frac{1}{J_{ii} - \sum_{k \neq i} J_{ik}^2 V_k}; \qquad \text{BP: } V_i = (J^{-1})_{ii}. \tag{B.4}$$

The AMP closure is the BP cavity with the exclusion of the receiving neighbour dropped ($V_{k\to i} \approx V_k$): an $O(1/N)$ change on a dense graph, an $O(1)$ change on a chain with two neighbours.

Theorem B.1 (The score is closure-independent)**.** *For a Gaussian posterior, the fixed-point condition for the means is the same linear system $Jm = h$ under mean field, BP, and AMP. All three schemes hence return the exact posterior mean and, via the score–posterior identity, the same exact joint score, for every (K, α, t), regardless of whether their variance closures are accurate or even solvable.*

The proof is one line: in each scheme the mean update at node i reads $m_i \leftarrow (h_i - \sum_{k\neq i} J_{ik} m_k)/J_{ii}$ at the fixed point; the closures modify the uncertainty bookkeeping, not the linear relation among means, and the unique solution of $Jm = h$ is the exact posterior mean. Numerically, AMP means agree with exact means to 4.3×10^{-12} and AMP scores with BP scores to 1.7×10^{-12} across all tested configurations.

For the variances the story splits. In the bulk the AMP closure (B.4) reads $V = 1/(J_d - 2\beta^2 V)$; the BP cavity is $V_{\text{cav}} = 1/(J_d - \beta^2 V_{\text{cav}})$. The entire difference between BP and AMP on the chain is a factor of two: both neighbours included where BP excludes one.

Theorem B.2 (AMP bulk variance, breakdown time, critical coupling)**.** *The AMP bulk closure is the quadratic $2\beta^2 V^2 - J_d V + 1 = 0$, with physical root*

$$V_{\text{AMP}} = \frac{J_d - \sqrt{J_d^2 - 8\beta^2}}{4\beta^2}, \tag{B.5}$$

which exists iff $J_d \geq 2\sqrt{2}\,|\beta|$, strictly stronger than BP's always-true $J_d > 2|\beta|$. Since $J_d(t)$ decreases monotonically in t, the existence condition fails, if ever, past a single time; solving it as an equality gives

$$\boxed{t_c(\alpha) = -\tfrac{1}{2}\log\frac{g}{1+g}, \qquad g(\alpha) = \frac{2\sqrt{2}\,|\alpha| - 1 - \alpha^2}{1-\alpha^2},} \tag{B.6}$$

finite exactly when $g > 0$, i.e.

$$\boxed{|\alpha| \;>\; \alpha_c \;=\; \sqrt{2}-1 \;\approx\; 0.4142.} \tag{B.7}$$

For $|\alpha| \le \alpha_c$ the AMP variance has a fixed point at every diffusion time; for $|\alpha| > \alpha_c$ it ceases to exist for all $t > t_c(\alpha)$. In the weak-coupling regime the two closures agree to second order, with

$$V_{\text{AMP}} - V_{\text{exact}} = \frac{2\beta^4}{J_d^5} + O\Big(\frac{\beta^6}{J_d^7}\Big) \;>\; 0: \tag{B.8}$$

AMP always overestimates the bulk variance, at fourth order in the coupling.

Toolbox B.1 — Roots, branch, and the phase boundary

Branch selection. The quadratic has roots $V_\pm = (J_d \pm \sqrt{J_d^2 - 8\beta^2})/(4\beta^2)$; as $\beta \to 0$, $V_- \to 1/J_d$ (the decoupled answer) while V_+ diverges, so V_- is the physical branch, and the attractive fixed point of the damped iteration used in practice. *Critical coupling.* $g > 0 \iff \alpha^2 - 2\sqrt{2}|\alpha| + 1 < 0 \iff |\alpha| \in (\sqrt{2}-1, \sqrt{2}+1)$; intersecting with $|\alpha| < 1$ leaves $|\alpha| > \sqrt{2}-1$. Consistency check at $t = \infty$: there the posterior is the bare prior, and the existence condition applied to Q_0 itself reads $(1+\alpha^2) \ge 2\sqrt{2}|\alpha|$, i.e. AMP applied to a bare AR(1) chain has a variance fixed point iff $|\alpha| \le \sqrt{2}-1$, which is (B.7) read backwards. *Weak coupling.* With $\epsilon := \beta^2/J_d^2$: $(1-4\epsilon)^{-1/2} = 1 + 2\epsilon + 6\epsilon^2 + O(\epsilon^3)$ for the exact variance; $1 - \sqrt{1-8\epsilon} = 4\epsilon + 8\epsilon^2 + O(\epsilon^3)$, divided by $4\epsilon J_d$, gives $V_{\text{AMP}} = (1 + 2\epsilon + 8\epsilon^2 + O(\epsilon^3))/J_d$; subtract. $\square$

Numerically, on a $K = 12$ chain at $\alpha = 0.8$: the AMP score error stays at 10^{-13}–10^{-12} for every t, while the variance error is 1.3×10^{-4} at $t = 0.05$, 3.5×10^{-2} at $t = 0.2$, and past $t \approx 0.23$ the variance update has no positive fixed point at all. The audit further verifies the closed form (B.5) against the converged iteration to 2.5×10^{-12}; the existence condition at all 180 points of an (α, t) scan; t_c against the bisected flip point of the iteration to 6.6×10^{-6} relative; convergence below α_c even at $t = 50$; and the weak-coupling ratio $(V_{\text{AMP}} - V_{\text{exact}})/(2\beta^4/J_d^5) \to 1$ within 0.4%. The interpretation is not that AMP is wrong; on a dense benchmark (random $N = 600$ symmetric positive-definite

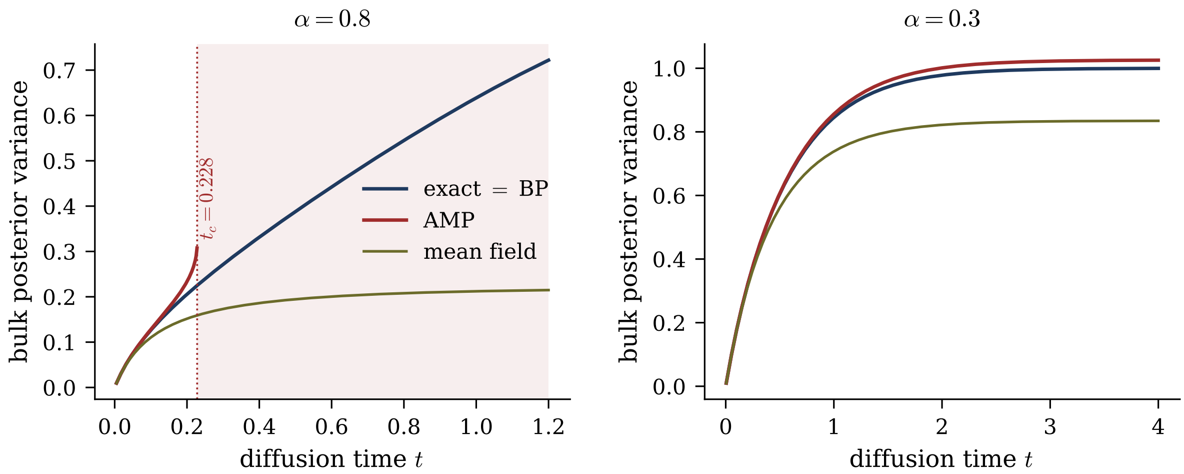

Figure B.1: The three closures in closed form. Left ($\alpha = 0.8 > \alpha_c$): exact = BP, AMP (B.5), and mean field; the AMP branch terminates with a square-root singularity at the breakdown time $t_c(0.8) = 0.228$ from (B.6) (shaded: no fixed point). Right ($\alpha = 0.3 < \alpha_c$): AMP exists at every t and slightly overestimates the variance, as predicted by (B.8).

precision) the same closure gets variances to 1.7×10^{-3} where mean field errs at 4×10^{-2}. It is built for a different graph: the central-limit justification of the per-node closure needs many incoming messages, and a chain provides two.

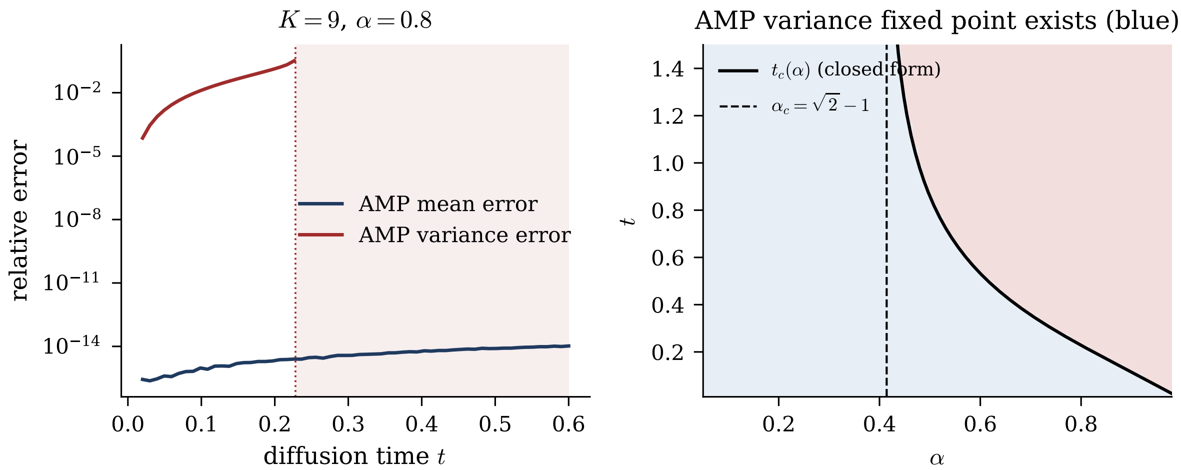

Figure B.2: BP against AMP on a finite chain ($K = 9$, $\alpha = 0.8$). Left: the AMP mean error (about 10^{-12}, flat in t: Theorem B.1) against the AMP variance error, which grows with t until the closure stops converging (shaded). Right: existence of the AMP variance fixed point in the (α, t) plane; the black curve is the boundary $t_c(\alpha)$ of (B.6), the dashed line $\alpha_c = \sqrt{2} - 1$; theory and iteration agree at all 180 scanned points.

Appendix C

Reproducibility

C.1 Repository layout

All source material lives in a single repository; the thesis directory is self-contained (LaTeX source, figures, and the figure script are committed together).

Path	Content
`thesis/`	this document: LaTeX source, figures, figure script
`research/gaussian-bp/`	Gaussian chain note, code, 72-check audit
`research/bp-from-scratch/`	independent BP derivation and implementation
`research/unified-note/`	consolidated working note, code, 55-check audit
`research/nongaussian-bp/`	grid BP vs Gaussian-projected BP experiments
`research/gaussian-ar1-bp/`	Gaussian AR(1) package (BP = matrix score)
`research/initial-experiments/`	early experiments and toy-model material

C.2 Audit protocol

Every closed-form claim is verified by a deterministic check against an independently computed reference (brute-force linear algebra, dense grid quadrature, or finite differences), collected in audit drivers that exit non-zero on any failure. The audit driver of `research/gaussian-bp` runs 72 checks: assembly and inversion of Σ_t; the score–posterior route against the matrix score; the $K = 2$ closed forms; Convention-A BP against the matrix posterior (means, variances, scores); bulk variance, covariance decay q^d, and the recombination identity; AMP mean and score closure-independence; the AMP variance formula, existence boundary, breakdown time, and weak-coupling law; the locality slopes; the band-fill coefficients (with the factorial variant explicitly rejected); spectral preservation and the large-t return. The audit driver of `research/unified-note` runs 55 checks on the consolidated note, including the Laplace $K = 1$ quadrature checks and the one-step boundary-message identity. `research/nongaussian-bp/` contains experiment drivers with logged master-error-identity residuals (10^{-15}–10^{-12} throughout), grid self-convergence gates, and per-trial CSV outputs; the grid calibration of Section 4.4 is experiment 01, the closure measurement of Section 4.5 experiment 02. All checks were re-executed on the build machine at the time of writing: 72/72 and 55/55 pass, and all experiment gates are green.

C.3 Figure-to-code map

Every figure is regenerated by one script, `figures/make_figures.py`, which recomputes each panel from the closed forms or quadrature of the corresponding chapter; the only figure that reads stored experimental data is Figure 4.2, built from the per-trial CSV of `nongaussian-bp` experiment 02. All figures are vector PDF with white background; randomness, where used, is seeded.

Figure	File	Source of the numbers
3.1	`fig_band_fill.pdf`	closed forms, Ch. 3
3.2	`fig_tridiag_loss.pdf`	closed forms, Ch. 3
3.3	`fig_local_vs_full.pdf`	closed forms, Ch. 3
4.1	`fig_laplace_k1.pdf`	quadrature, Ch. 4
4.2	`fig_laplace_closure.pdf`	`nongaussian-bp` exp. 02 CSV
B.1	`fig_bulk_variance.pdf`	closed forms, App. B
B.2	`fig_bp_vs_amp.pdf`	closed forms + iteration, App. B

C.4 Environment

Python ≥ 3.10 with NumPy, SciPy, and Matplotlib; no GPU, no neural network framework, no stochastic training. All experiments are deterministic given the recorded seeds; the error measures of the locality and truncation studies are seed-free by construction (closed-form traces). The document compiles with a single run of `tectonic main.tex`.

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